



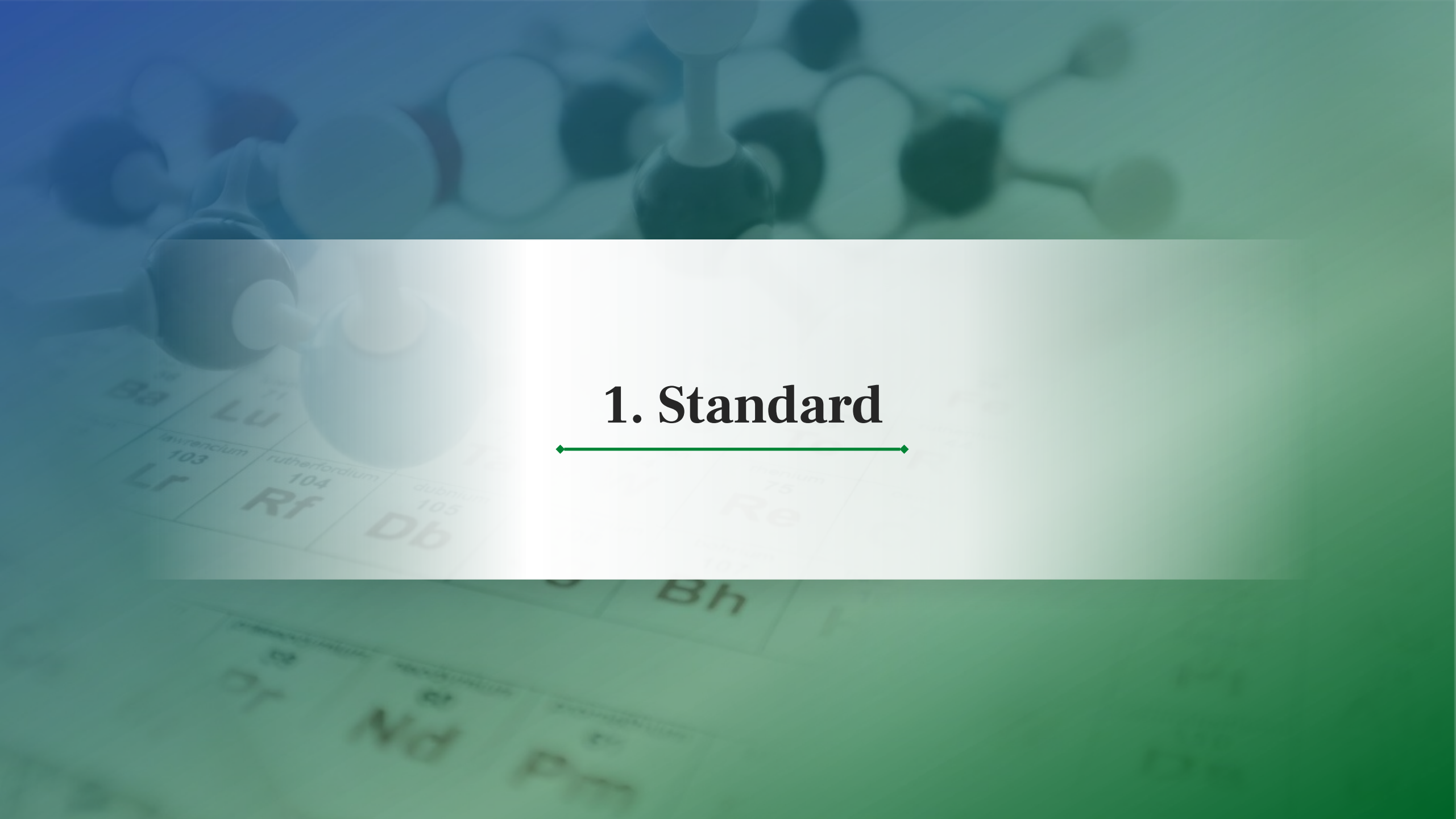
Chapter 12

Data standardization



Contents

- 1. Standard**
- 2. Calibration transfer methods**
- 3. AI for data standardization**



1. Standard



**LabRam HR
Evolution**



**Handheld
Raman
Instrument**



NI



**Handheld
NIR
Instrument**
«»



**500MHz
NMR**



**Benchtop
NMR**



Standard

In recent two decades, China has promulgated 80 national, industrial, and local standards, involving chemical, food, agriculture, textile, and other fields. The relevant **NIR** standard methods from different levels in China are as follows:

1. GB/T 18,868-2002 Method for determination of moisture, crude protein, crude fat, crude fiber, lysine, and methionine in feeds—Near-infrared reflectance spectroscopy method
2. GB/T 12,008.3-2009 Plastics-Polyether polyols—Part 3: Determination of hydroxyl number
3. GB/T 24,895-2010 Inspection of grain and oils—General regulations for model authentication of near-infrared analysis and administration and maintenance of network
4. GB/T 25,219-2010 Inspection of grain and oils—Determination of starch content in maize—Near-infrared spectroscopy method
5. GB/T 24,900-2010 Inspection of grain and oils—Determination of moisture content in maize—Near-infrared spectroscopy method
6. GB/T 24,902-2010 Inspection of grain and oils—Determination of crude fat content in maize—Near-infrared spectroscopy method
7. GB/T 24,896-2010 Inspection of grain and oils—Determination of moisture content in paddy—Near-infrared spectroscopy method
8. GB/T 24,897-2010 Inspection of grain and oils—Crude protein determination in rice—Near-infrared spectroscopy method
9. GB/T 24,898-2010 Inspection of grain and oils—Determination of moisture content in wheat—Near-infrared spectroscopy method
10. GB/T 24,899-2010 Inspection of grain and oils—Determination of crude protein in wheat—Near-infrared spectroscopy method
11. GB/T 24,871-2010 Inspection of grain and oils—Crude protein determination in wheat flour—Near-infrared spectroscopy method
12. GB/T 24,872-2010 Inspection of grain and oils—Determination of ash content in wheat flour—Near-infrared spectroscopy method
13. GB/T 24,870-2010 Inspection of grain and oils—Crude protein and crude fat determination in soybean—Near-infrared spectroscopy method
14. GB/T 29,858-2013 Standard guidelines for molecular spectroscopy multi-variate calibration quantitative analysis
15. GB/T 34,406-2017 Identification of pearl powder—Near-infrared spectroscopy method
16. GB/T 36,691-2018 Methyl vinyl silicone rubber—Determination of vinyl content—Near-infrared spectroscopy
17. GB/T 37,969-2019 Standard guidelines for near-infrared qualitative analysis
18. GB/T 7383-2020 Non-ionic surface active agents—Determination of hydroxyl value
19. GB/T 13,892-2020 Surface active agents—Determination of iodine value
20. ChP 2015 The Pharmacopoeia of the People's Republic of China (2015) 9104 Guidelines for Near-Infrared Spectrophotometry
21. NY/T 1423-2007 Method for Quick Discrimination of Meat and Bone Meal in Fishmeal and Ruminant Concentrate Supplement—Near-infrared reflectance spectroscopy method
22. NY/T 1841-2010 Non-destructive determination of soluble solid and titratable acidity in apple fruit by near-infrared spectroscopy method
23. NY/T 2797-2015 Non-destructive determination of fat in meat by near-infrared spectroscopy method
24. NY/T 2794-2015 Determination of amino acids content in peanut—Near-infrared spectroscopy method
25. NY/T 3105-2017 Determination of oil content in vegetable oilseeds—Near-infrared spectroscopy method
26. NY/T 3299-2018 Determination of oleic acid and linoleic acid in vegetable oilseeds—Near-infrared spectroscopy method
27. NY/T 3298-2018 Determination of crude protein content in vegetable oilseeds—Near-infrared spectroscopy method
28. NY/T 3297-2018 Determination of total phenolic compounds and tocopherols in rapeseed seeds—Near-infrared spectroscopy method
29. NY/T 3295-2018 Determination of erucic acid and glucosinolate in rapeseed—Near-infrared spectroscopy method
30. NY/T 3512-2019 Non-destructive determination of protein in meat—Near-infrared spectroscopy method
31. NY/T 3679-2020 Technical code of practice for screening high oleic acid peanut—Near-infrared spectroscopy method
32. SN/T 3896.1-2014 Quantitative analysis of fiber in textiles for import and export—Near-infrared spectroscopy method—Part 1: Mixture of polyester fiber and cotton fiber
33. SN/T 3896.2-2015 Quantitative analysis of fiber in textiles for import and export—Near-infrared spectroscopy method—Part 2: Mixture of polyester fiber and polyurethane fiber
34. SN/T 3896.3-2015 Textiles for import and export—Quantitative analysis of fiber—Near-Infrared spectroscopy method—Part 3: Mixture of polyamide fiber and polyurethane fiber
35. SN/T 3896.4-2015 Textiles for import and export—Quantitative analysis of fiber—Near-Infrared spectroscopy method—Part 4: Mixture of cotton fiber and polyurethane fiber
36. SN/T 3896.5-2015 Textiles for import and export—Quantitative analysis of fiber—Near-infrared spectroscopy method—Part 5: Mixture of polyester fiber and rayon fiber
37. SN/T 3896.6-2017 Textiles for import and export—Quantitative analysis of fiber—Near-infrared spectroscopy method—Part 6: Mixture of polyester fiber and wool fiber
38. SN/T 3896.7-2020 Quantitative analysis of fiber in textiles for import and export—Near-infrared spectroscopy method—Part 7: Mixture of polyester fiber and polyamide fiber
39. SN/T 3896.8-2020 Fiber quantitative analysis of textile for import and export—Near-infrared spectroscopy —Part 8: Mixture of cotton and polyamide fiber
40. SN/T 5233-2020 Import and export textile material test of moisture regain of raw cotton—Near-infrared spectroscopy method
41. SB/T 11,149-2015 Technical specifications of waste plastics collection and sorting
42. FZ/T 01,144-2018 Textiles—Quantitative analysis of fiber—Near-infrared spectroscopy method
43. FZ/T 01,150-2019 Textile -Test method for identification of bamboo fibre and viscose from bamboo—Near-infrared spectroscopy method
44. LY/T 2151-2013 Method for determination of holocellulose and acid-insoluble lignin in wood—Near-infrared spectroscopy method
45. LY/T 2053-2012 Standard method for near-infrared qualitative analysis of wood
46. GH/T 1260-2019 Method for Moisture, Total Polyphenols, and Caffeine in Instant Tea in Solid Form—Near-infrared reflectance spectroscopy method
47. GH/T 1259-2019 Method for moisture, total polyphenols, and caffeine in tea polyphenol products—Near-infrared reflectance spectroscopy method
48. QB/T 2812-2006 Paper-online determination of weight and moisture (The near-infrared spectroscopy method)
49. HG/T 3505-2020 Surface active agents—Determination of saponification value
50. DB12/T 347-2007 Rapid method for determination of crude protein in wheat and corn by near-infrared spectroscopy
51. DB22/T 1605-2012 Rapid and nondestructive detection of ash content, moisture, water-insoluble solids, water-saturated butanol extract in the Ginseng—Near-infrared spectroscopy method
52. DB32/T 2269-2012 Determination of Cottonseed Oil Content by Near-Infrared Spectroscopy
53. DB21/T 2048-2012 Determination of crude protein, crude fat, crude fibre, moisture, calcium, Sphosphorus, crude ash, water-soluble chlorides, amino in feeds Near-infrared reflectance spectroscopy
54. DB22/T 1812-2013 Rapid and nondestructive detection of polysaccharides in the Ginseng-Near-infrared spectroscopy method
55. DB53/T 497-2013 Guidelines for the establishment and validation of near-infrared calibration models for the main chemical constituents of tobacco and tobacco products
56. DB53/T 498-2013 Determination of main chemical components in tobacco and tobacco products—Near-infrared diffuse reflectance spectroscopy method

Standard



Standard	Main terms defined related to Raman spectroscopy	
ASTM E131-10 (2015) ⁸ (Standard Terminology Relating to Molecular Spectroscopy)	Anti-Stokes line Background Baseline Beamsplitter Certified reference material Fiber optics Filter Fluorescence Luminescence Monochromator Multiple correlation coefficient Numerical aperture (NA) Optical fiber Precision Principal component analysis Raman line (band) Visible Wavelength Wavenumber	Raman shift Raman spectrum Reference material (ISO Guide 30) Reference spectrum Resolution Sample pathlength Sample spectrum Scattering, 90° (or 180°) Signal-to-noise ratio Spectral slit width Spectrometer Spectroscopy Spectrum Standard error of calibration Stokes line (band) Ultraviolet Visible Wavelength Wavenumber
ASTM E2642-09 (2015) ⁹ (Standard Terminology for Scientific Charge-Coupled Device (CCD) Detectors)	Advanced inverted mode operation (AIMO) Analog-to-digital (A/D) converter Anti-blooming structure Antireflective (AR) coating Back-illuminated CCD Binning CCD bias Charge Charge-coupled device Charge transfer Charge transfer efficiency Column Complementary metal oxide semiconductor Correlated double sampling Cosmic event Dark current Dark noise Deep depletion CCD Dynamic range Electron-multiplying CCD Etaloning Exposure time Frame Frame-transfer CCD Front-illuminated CCD Full-frame CCD Full well capacity Gate structure Horizontal binning Indium tin oxide	Intensified CCD Interline transfer CCD Linear array CCD Multi-pinned phase (MPP) Noise Open electrode CCD Horizontal register Output amplifier Output node Parallel shift Peltier cooler Pixel Pixel non-uniformity Potential well Quantum efficiency (QE)) Read noise Readout rate Region of interest (ROI) Row Serial register Shot noise limit Signal-to-noise ratio (SNR) Silicon Slow-scan CCD Spectral rate Thermoelectric cooling Thinning Vertical binning
ASTM D8333 ¹⁰ (Standard Practice for Preparation of Water Samples with High, Medium, or Low Suspended Solids for Identification and Quantification of Microplastic Particles and Fibers Using Raman Spectroscopy, IR Spectroscopy, or Pyrolysis–GC/MS)	Effluent Influent Microplastic Surface waters Suspended solids	

Standard	Main terms defined related to Raman spectroscopy	
ASTM D6122 ¹¹ (Standard Practice for Validation of the Performance of Fluorescent Online, At-Line, Field and Laboratory Infrared Spectrophotometer, and Raman Spectrometer Based Analyzer Systems)	Action limit Analyzer Analyzer calibration Analyzer model Analyzer repeatability Analyzer result Analyzer site precision Analyzer validation status Analyzer validation test Between-method reproducibility Chart expectations based on the multivariate model Control limits Exponentially weighted moving average control General validation Individual observation control chart In-line detection methods In-line probe Instrument Instrument standardization Line sample Local validation Model degrees of freedom (DOF) Model variables Moving range of two control chart Multivariate calibration Multivariate model Calibrated detector Calibrated optical radiation source Calibration Certified reference material (E131) Certified value Diffuse scatterer Flux Grating equation Inner filter effects Lambertian reflector Line of detection Noise level Photobleaching Analysis Calibration Calibration model Calibration samples Estimate Model variation Multivariate calibration Acceptance Accepted reference value Accuracy Deviation Error of result Method of least squares Observed value Precision Repeatability Reproducibility Robustness Spectrometer/spectrophotometer qualification Surrogate calibration Surrogate test method Accumulations Charge-coupled device Charge-coupled device binning Exposure time Incident light Raman scattering effect	Nearest neighbor distance inlier Optical background Optical reference filter Outlier detection limits Outlier detection methods Outlier spectrum Performance test Physical correction Post-processing Predicted primary test method result Prediction deviation Pre-processing Primary test method (PTM) Predicted primary test method result (PTMR) Process analyzer system Process analyzer validation samples Spectrometer cell, Raman Standard error of calibration Test sample Transfer optics Uncertainty of predicted primary test method result (UPTPMR) Validated result Validation reference material (VRM) Validation samples Validation test Qualification Quantum counter Raman scattering Rayleigh scattering Responsivity Sensitivity Spectral bandwidth/bandpass or resolution Spectral flux Spectral responsivity Spectral slit width (E131) Traceability Transfer standard Transition dipole moment Reference method Reference values Spectrometer/spectrophotometer Surrogate calibration Surrogate method Validation samples Sample (size) Specification/tolerance limits (within laboratory) standard deviation Statistical Systematic error of result Systematic sampling Test method/result/specimen, etc. Uncertainty Unit Variance Raman spectroscopy Raman spectrum Scattered light Signal strength Wavenumber
ASTM E2719 ¹² (Standard Guide for Fluorescence—Instrument Calibration and Qualification)		
ASTM E1655 ¹³ (Standard Practices for Infrared Multivariate Quantitative Analysis)		
ASTM E456 ¹⁴ (Standard Terminology Relating to Quality and Statistics)		
ASTM E2056 ¹⁵ (Standard Practice for Qualifying Spectrometers and Spectrophotometers for Use in Multivariate Analyses, Calibrated Using Surrogate Mixtures)		
ASTM D7940 ¹⁶ (Standard Practice for Analysis of Liquefied Natural Gas (LNG) by Fiber-Coupled Raman Spectroscopy)		

Standard	Main terms defined related to Raman spectroscopy	
ISO/TS 80004-13:2017 ^{17, 18} (Nanotechnologies—Vocabulary—Part 13: Graphene and related two-dimensional (2D) materials) ISO/TS 80004-6:2021 ¹⁹ (Nanotechnologies—Vocabulary—Part 6: Nano-object characterization) ISO 18115-1:2013 ²⁰ (Surface chemical analysis—Vocabulary—Part 1: General terms and terms used in spectroscopy) ISO 18115-2:2013 ²² (Surface chemical analysis—Vocabulary—Part 2: Terms used in scanning-probe microscopy) ISO/DIS 18115-3 (under development) ²⁴ (Surface chemical analysis—Vocabulary—Part 3: Terms used in optical interface analysis) ISO 23978:2020 ²¹ (Natural Gas—Upstream area—Determination of composition by Laser Raman spectroscopy)	Raman spectroscopy Fluorescence Raman effect Raman spectroscopy Data preprocessing Multivariate analysis, MVA Multivariate curve resolution, MCR Reference material, RM Reference material, certified, CRM Relative resolution of a spectrometer Apertureless Raman microscopy Anti-Stokes scattering Aperture Fluorescence quenching Image/Map Near-field Raman microscopy Numerical aperture (NA) Optical resolution Anti-Stokes Raman scattering Background signal (ISO 18115 1: 2103) Backscattering configuration CCD detector Circular polarization Coherent anti-Stokes Raman scattering (CARS) Coherent Raman scattering microscopy (CRS) Coherent Stokes Raman scattering (CSRS) Confocal optical microscopy (ISO/TS 80004 6:2021) Confocal Raman microscopy Confocal volume (ISO 10934(en): 2020) Depolarization ratio Depth of focus (ISO 19262:2015) Edge filter Electronic Raman scattering Enhancement factor Fluorescence (ISO 18115-2:2013) Fluorophore (ISO 18115 2:2013) Hyper-Raman scattering Irradiance (ISO 29444:2017) Luminescence (IEC 60050 845: 1987) Monochromator (ISO 17861:2014) Monochromatic radiation (ISO 10934: 2020) Notch filter Objective lens (ISO 9849:2000) Peak height (ISO 7941:1988) Peak shape Peak width (ISO 18115 1:2013) Photobleaching Photodetector Photomultiplier tubes, photomultipliers (PMTs) Plasmon Polarizability Polarization Polarized Raman spectroscopy Raman effect Multichannel photodetector Avalanche photodiode (APD)	Surface-enhanced Raman spectroscopy (SERS) Signal-to-noise ratio Tip-enhanced Raman spectroscopy (TERS) Resolution, lateral Sample area viewed by the analyzer Signal-to-noise ratio Smoothing Samples “multivariate analysis” Variables “multivariate analysis” Polarization Raman effect Raman spectroscopy Stokes scattering Surface-enhanced Raman scattering (SERS) Surface-enhanced resonant Raman scattering (SERRS) Tip-enhanced Raman spectroscopy (TERS) Tip enhancement Polarizer Planar-linear polarization Raman amplification Raman depth profiling Raman mapping/imaging Raman optical activity (ROA) Raman scattering effect Raman spectroscopy (ISO 80004 13: 2017) Raman tensor Raman shift Rayleigh scattering Resonance Resonance Raman spectroscopy (RR) Resonant excitation profile Rotational Raman spectroscopy Selection rules Silicon diode detector Spontaneous Raman spectroscopy Spatially offset Raman spectroscopy (SORS) Spectral purity Stimulated Raman scattering (SRS) Stokes Raman scattering Surface-enhanced Raman spectroscopy (SERS) Surface-enhanced Raman scattering (SERS) (ISO 18115 2: 2013) Surface-enhanced resonant Raman scattering Surface-enhanced resonant Raman spectroscopy (SERRS) (ISO 18115 2:2013) Transmission Raman spectroscopy (TERRS) Transmission configuration Vibrational-rotational spectroscopy Wavelength, λ (ISO 80000 3:2004) Wavenumber, ν (ISO 19702:2015) Signal intensity External cavity/interactivity ² Base span calibration

Standard	Main terms defined related to Raman spectroscopy	
ISO Guide 30:2015 ^{25, 22} (Reference materials—Selected terms and definitions)	Certified reference material (CRM) Certified value Characterization Homogeneity Interlaboratory comparison/study/ test, collaborative study	Lifetime Reference material (RM) Reference method/procedure Sample Stability

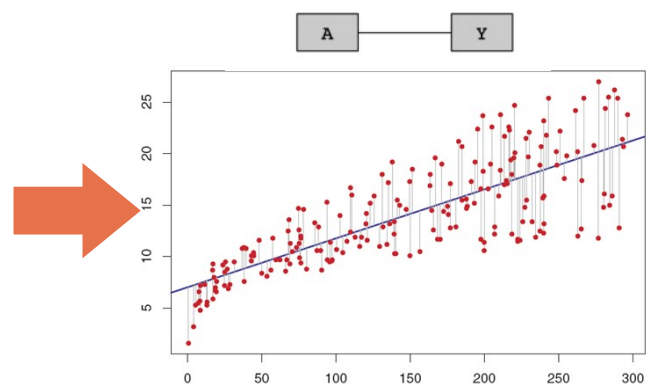
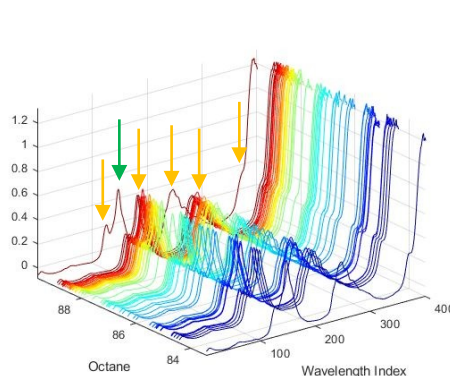
²⁵Standards not further commented on in this review.

¹⁸ISO/TS 80004-13:2017 Nanotechnologies—Vocabulary—Part 13: Graphene and related two-dimensional (2D) materials currently published is to be replaced by ISO/AWI TS 80004-13

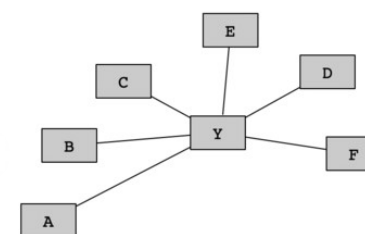
²Measuring gas with a spectrograph outside and inside the laser, respectively.

Multiple correction method

Multivariate Linear Regression, MLR
Principal Component Regression, PCR
Partial Least Squares Regression, PLSR



One-variable linear regression



MLR

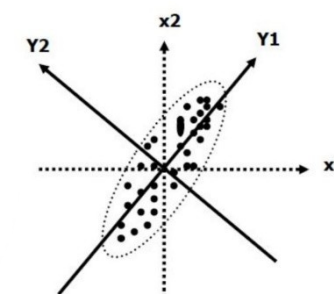
- Linear model
- Quantitative analysis;
- Special system;

Poor linear correlation
with the target

PLSR

PCR

Information
Redundancy

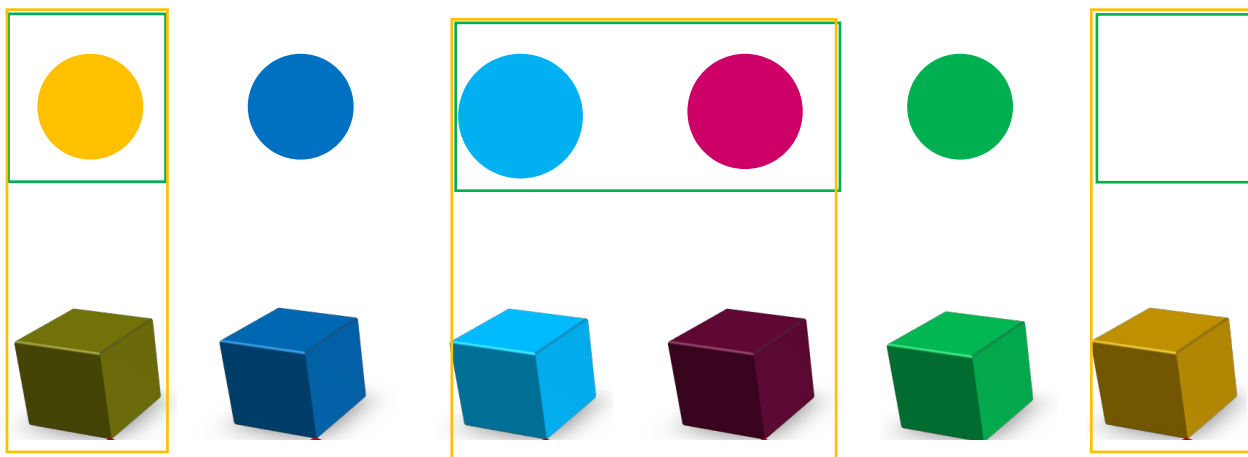


Principal component analysis
PCA

Samples Select

Kennard – Stone algorithm (KS)

$$d_x(p, q) = \sqrt{\sum_{j=1}^N [x_p(j) - x_q(j)]^2}; p, q \in [1, N]$$



Sample Set Partitioning based on Joint X-Y Distance (SPXY)

$$d_x(p, q) = \sqrt{\sum_{j=1}^N [x_p(j) - x_q(j)]^2}; p, q \in [1, N]$$

$$d_y(p, q) = \sqrt{(y_p - y_q)^2} = |y_p - y_q|; p, q \in [1, N]$$

$$d_{xy}(p, q) = \frac{d_x(p, q)}{\max_{p, q \in [1, N]} d_x(p, q)} + \frac{d_y(p, q)}{\max_{p, q \in [1, N]} d_y(p, q)}; p, q \in [1, N]$$

https://blog.csdn.net/qq_43816569

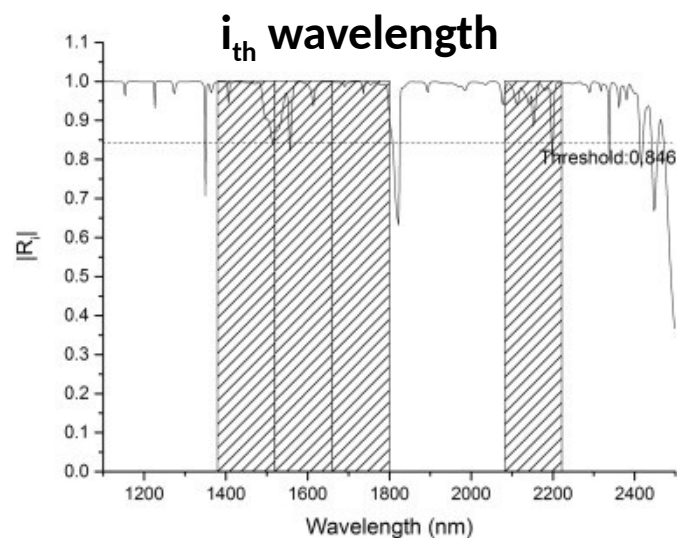
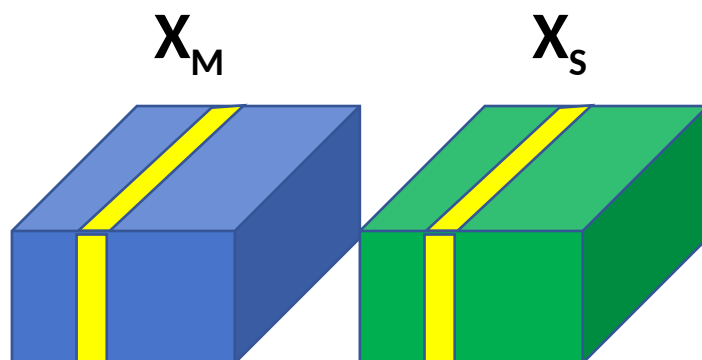
**Spectroscopic
Data X**

**Physical Property
Data(Y)**

- KS only considers the spatial distribution of spectral data.
- SPXY considers the distance between two spatial data points.

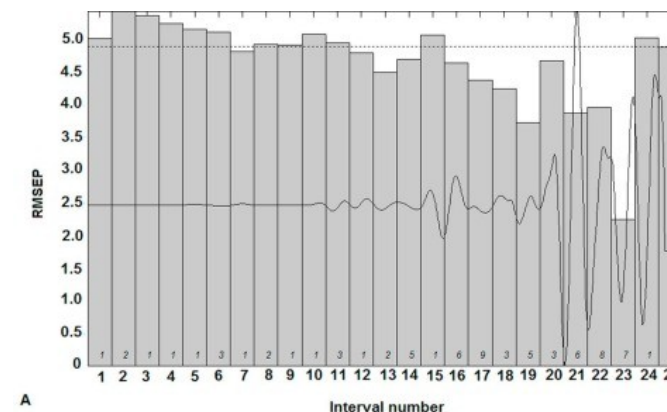
Wavelength Select

Correlation analysis wavelength selection method (CAWS)

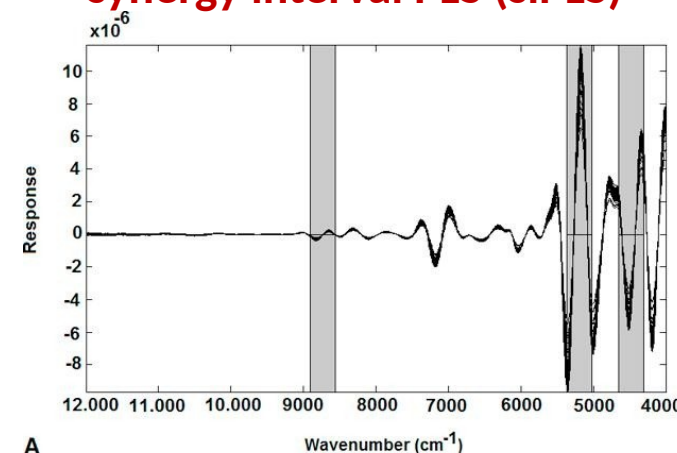


interval PLS (iPLS)

moving window PLS (mwPLS)

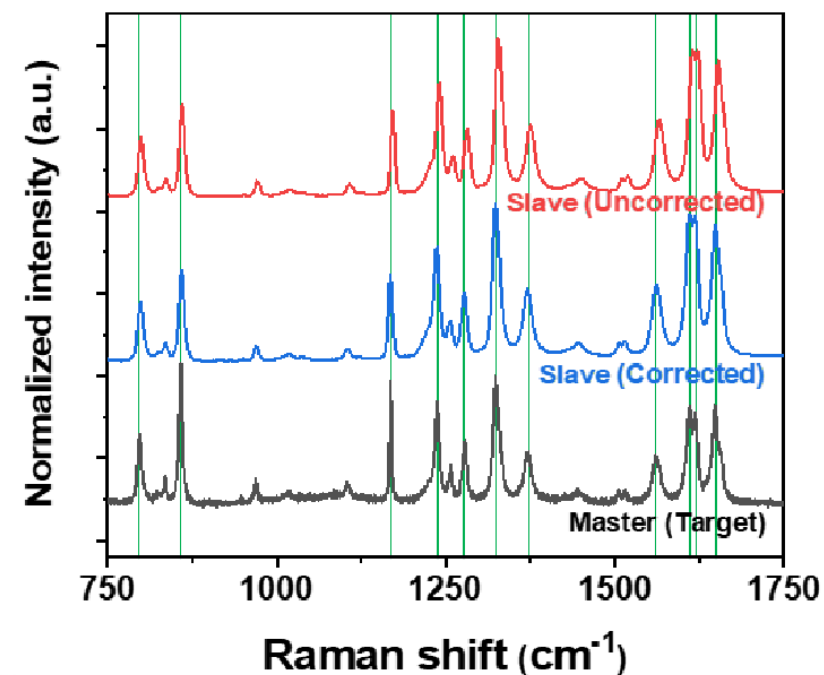
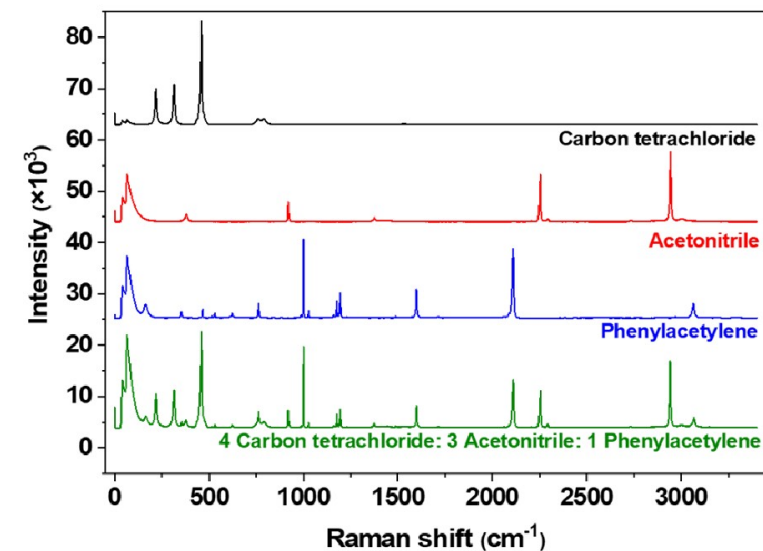
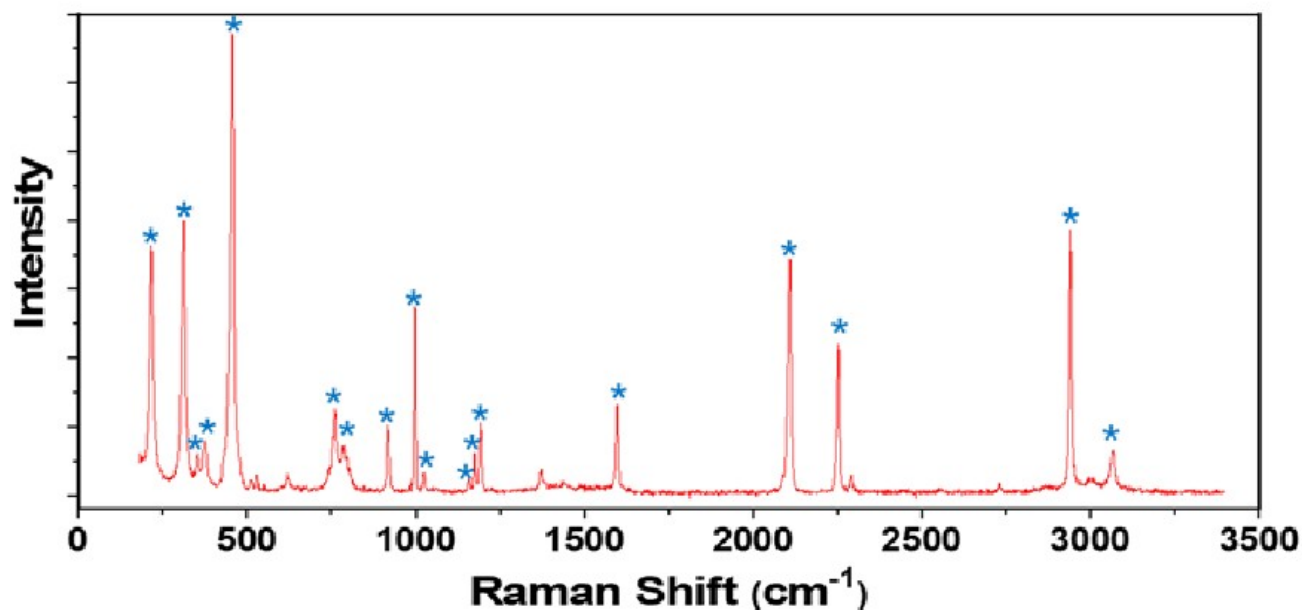


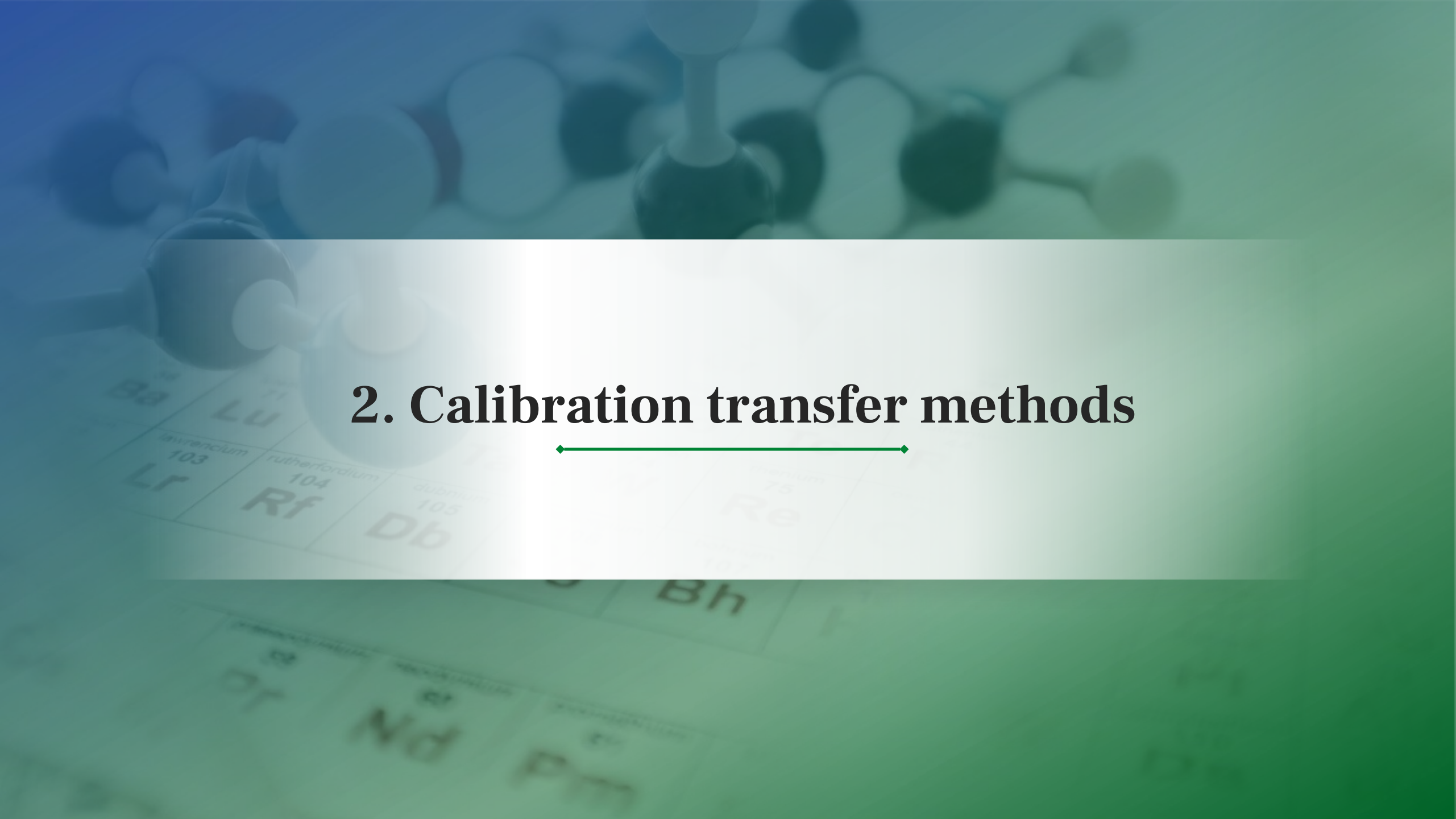
synergy interval PLS (siPLS)



Standard sample

Compound	Source
Naphthalene	Mallinckrodt ⁴
1,4-Bis(2-methylstyryl)benzene (a laser dye)	Aldrich ⁵
Sulfur	Aldrich ⁵
50/50 (v/v) toluene/acetonitrile	Mallinckrodt ⁴
4-Acetamidophenol ⁶	Aldrich ⁵
Benzonitrile	Aldrich ⁵
Cyclohexane	Mallinckrodt ⁴
Polystyrene	Aldrich ⁵





2. Calibration transfer methods

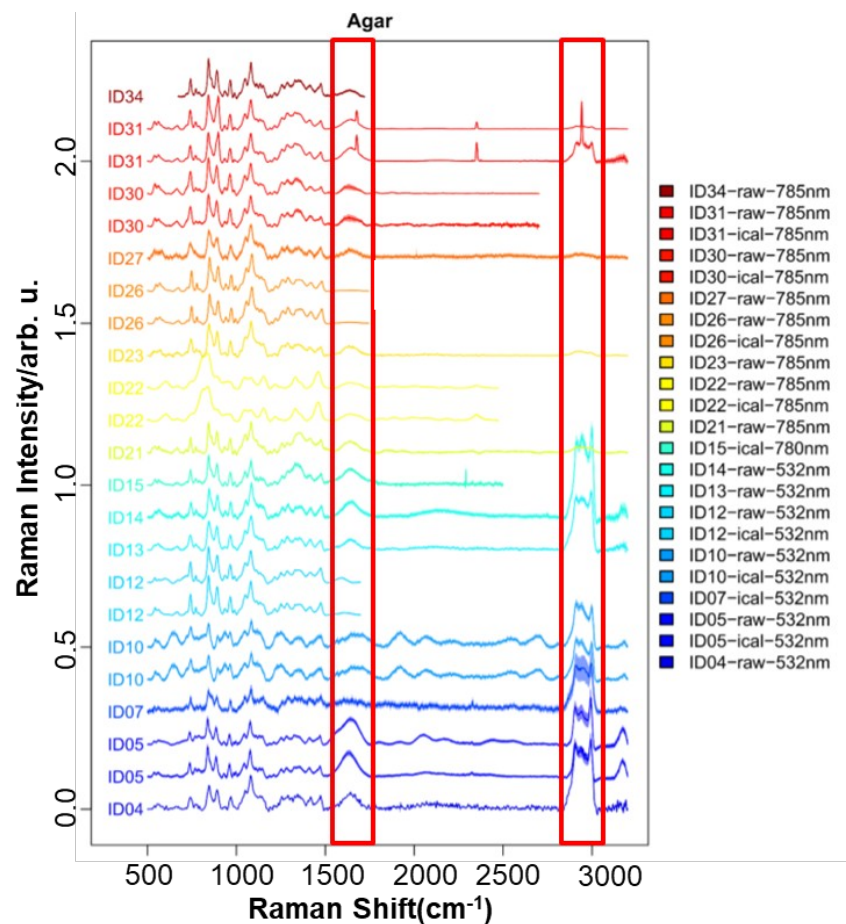
Introduction

A situation is often encountered in the process of spectral analysis that when the model established on one spectrometer (**master**, primary, parent instrument) is used on another spectrometer (**slave**, child instrument), the model cannot give correct prediction results due to the difference in spectra measured by different instruments. The **first thing** to solve this problem is to improve the standardization of instrument hardware processing, improve the level of processing technology, and reduce the differences between the master and the slaves in terms of devices. Instrument standardization makes the spectra from the same samples measured by different instruments as consistent as possible. There have been many reports on the standardization of spectroscopic instrument hardware such as calibrating the wavelength accuracy, absorbance accuracy, resolution, spectral response line type, and symmetry of the spectrometer through sharp-line emission light source, standard materials, etc. As well, there have been developments on instruments by optimizing optical components, assembly processes, and control strategies with performance indicators within the allowable range of variation.

This is called the **First Principle of instrument calibration**, which is the most fundamental basis of modern spectroscopic analysis technology.

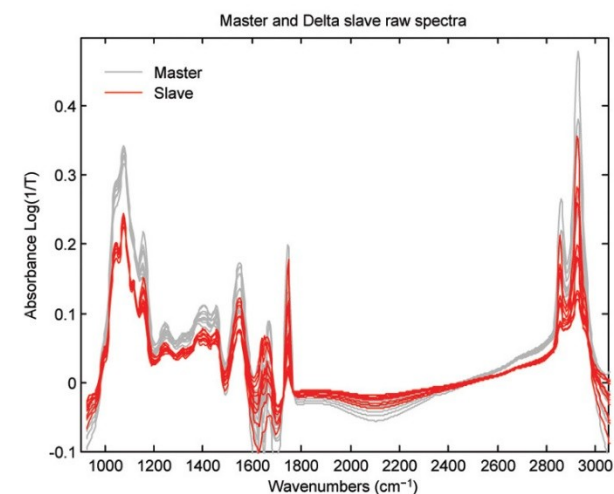
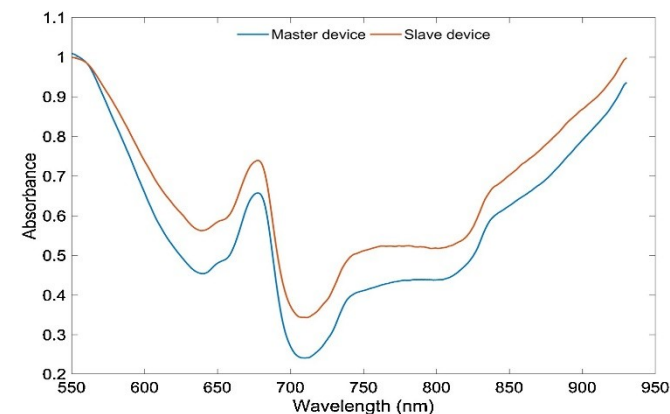
Instrument differences

Raman (Agar)



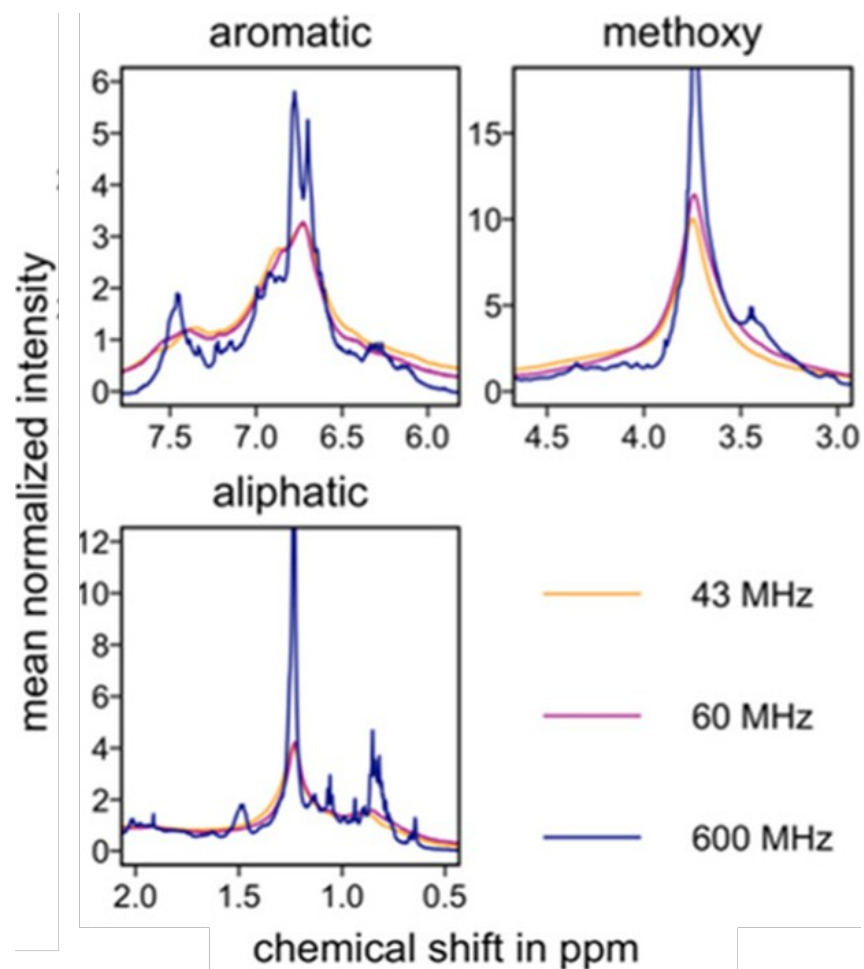
T. Bocklitz et al. *Anal. Chem.* **2020**, 92, 15745–15756

NIR(Apples, Milk)



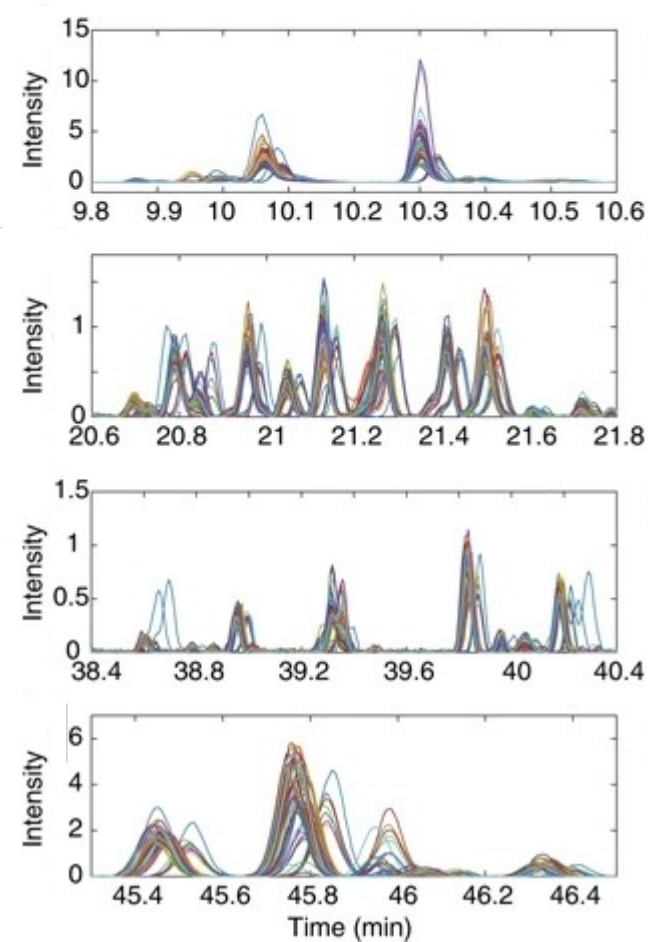
F. Dehareng group. *J. Dairy Sci.* **2015**, 98, 4, 2150–2160

NMR (lignin)



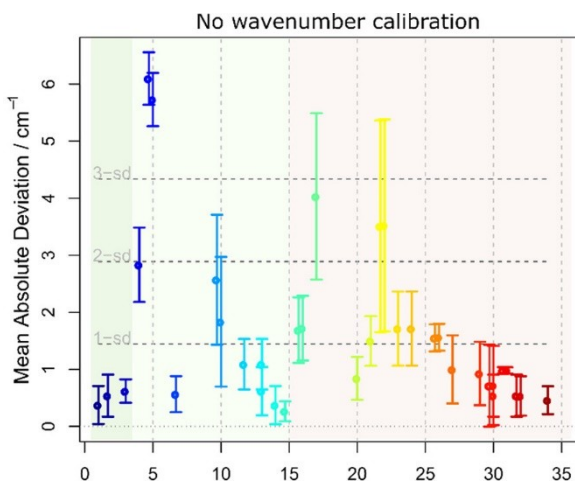
Y. B. Monakhova et al. *Anal. Chem.* **2022**,94,9,3997-4004

Chromatograph

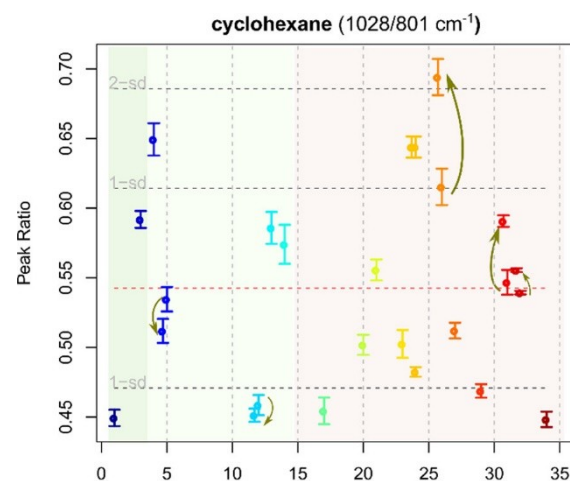


Q. X. Zheng et al. *Sci. Rep.* **2017**,7,256

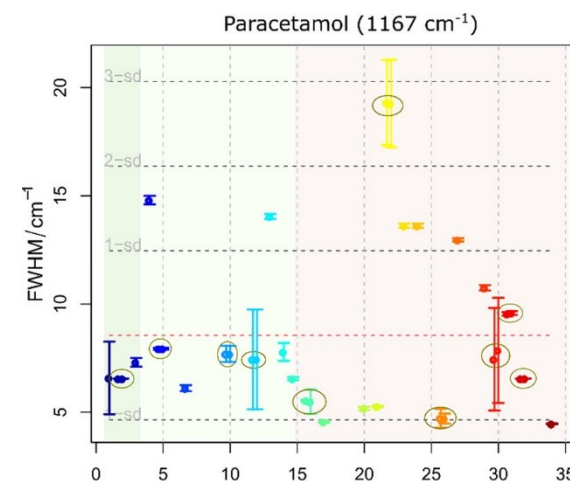
■ ID01-raw-514nm	■ ID06-raw-532nm	■ ID11-ical-532nm	■ ID16-raw-785nm	■ ID22-raw-785nm	■ ID27-raw-785nm	■ ID31-raw-785nm
■ ID02-raw-514nm	■ ID07-ical-532nm	■ ID12-ical-532nm	■ ID17-raw-785nm	■ ID23-raw-785nm	■ ID28-raw-785nm	■ ID32-ical-785nm
■ ID02-ical-514nm	■ ID08-raw-532nm	■ ID12-raw-532nm	■ ID18-raw-785nm	■ ID24-ical-785nm	■ ID29-raw-785nm	■ ID32-raw-785nm
■ ID03-raw-515nm	■ ID09-ical-532nm	■ ID13-raw-532nm	■ ID19-raw-785nm	■ ID24-raw-785nm	■ ID30-raw-785nm	■ ID33-raw-785nm
■ ID04-raw-532nm	■ ID10-ical-532nm	■ ID14-raw-532nm	■ ID20-raw-785nm	■ ID26-ical-785nm	■ ID30-ical-785nm	■ ID34-raw-785nm
■ ID05-ical-532nm	■ ID10-raw-532nm	■ ID15-ical-780nm	■ ID21-raw-785nm	■ ID26-raw-785nm	■ ID31-ical-785nm	■ ID35-raw-785nm
■ ID05-raw-532nm	■ ID11-raw-532nm	■ ID16-ical-785nm	■ ID22-ical-785nm			



Mean absolute deviation of acetaminophen on different instruments



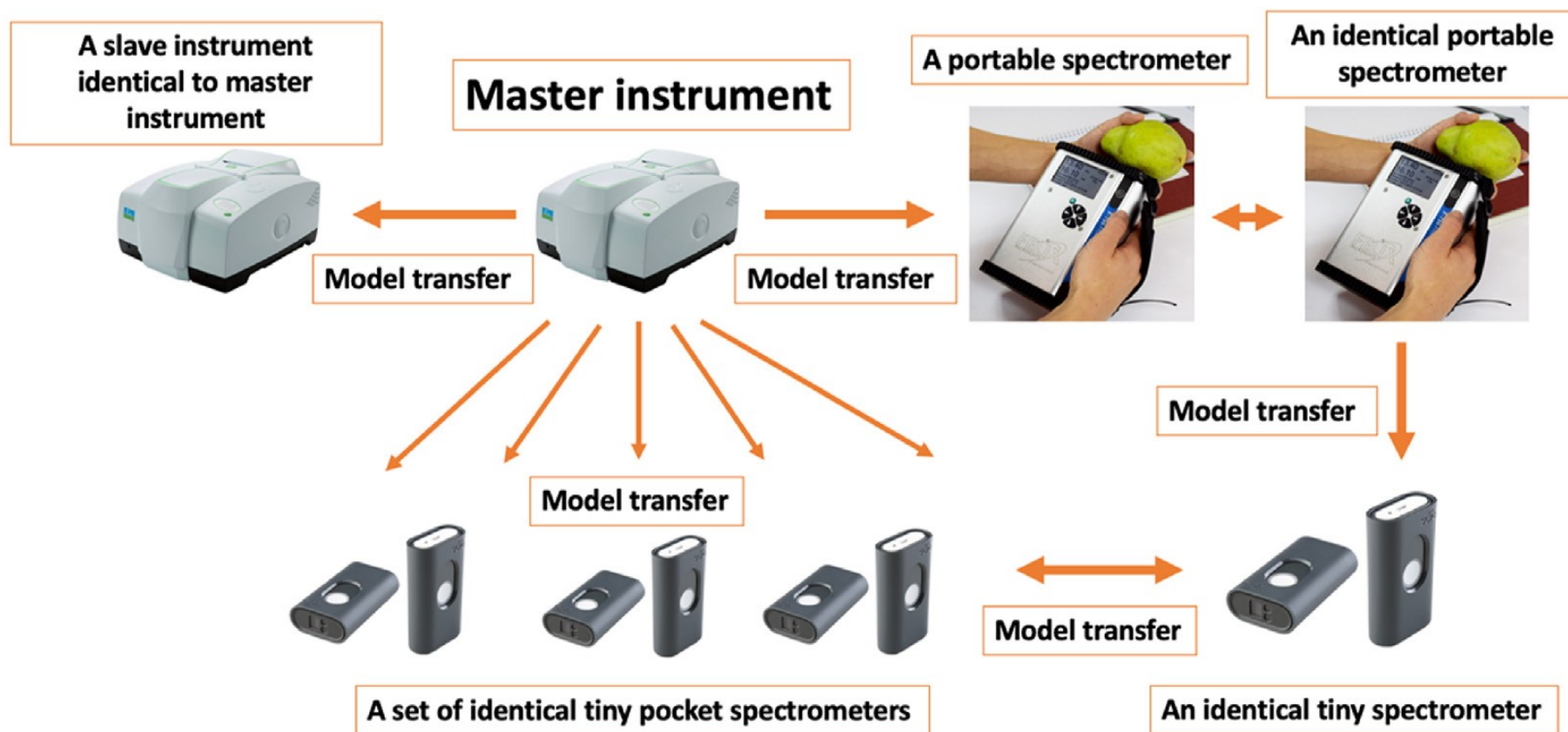
Ratio of two characteristic peak-to-peak strength of cyclohexane on different instruments



Acetaminophen 1167cm⁻¹ characteristic peak of FWHM on different instruments

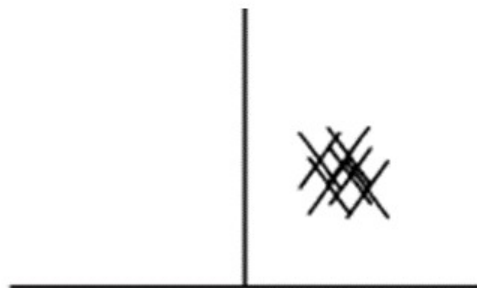
➤ Signals can be divided into peak position, peak strength and half peak width correction.

Master and Slaver Instruments



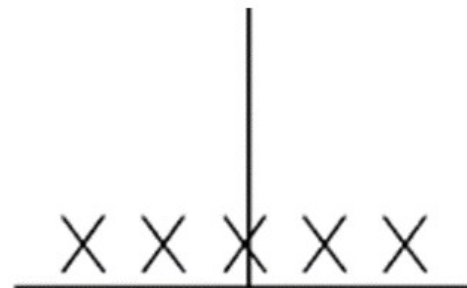
Accuracy and Precision

Reference value



Precise but not accurate

Reference value



Accurate but not precise

Reference value



Neither accurate nor precise

Reference value



Precise and accurate

Significance testing

Precision – F test

$H_0:$

Evaluation parameters	Condition
R	Similar magnitude to original calibration model
RMSE	Similar magnitude to original calibration model
Bias	Not significant($\text{Bias} < \text{Bias}_{\text{max}}$)
HQI	[0,1] Spectral consistency
.....	

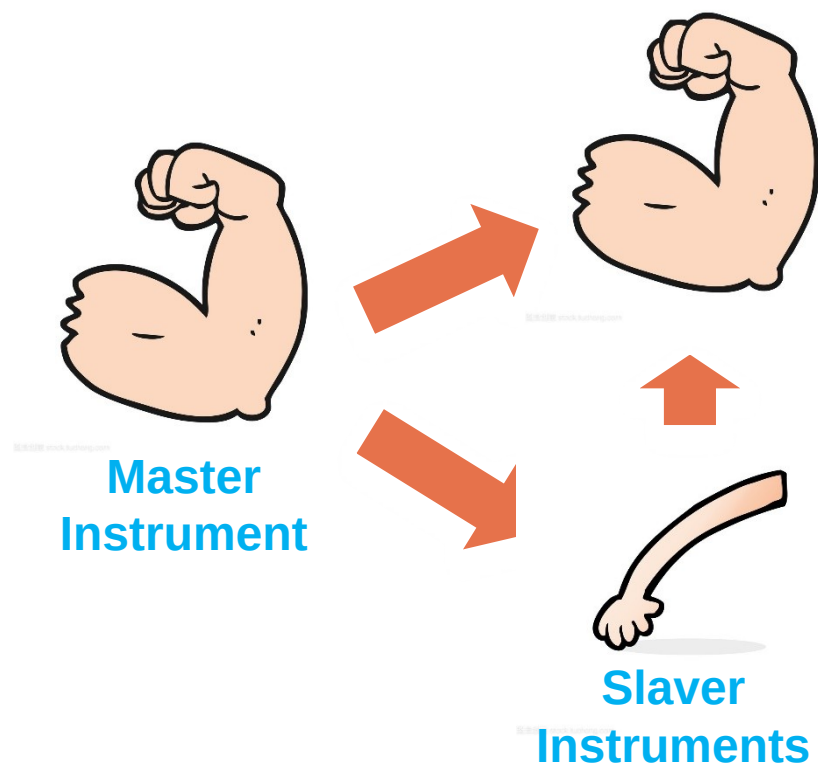
$$F < F_{v_1, v_2, \alpha} \quad \text{Accept}$$

Accuracy – t test

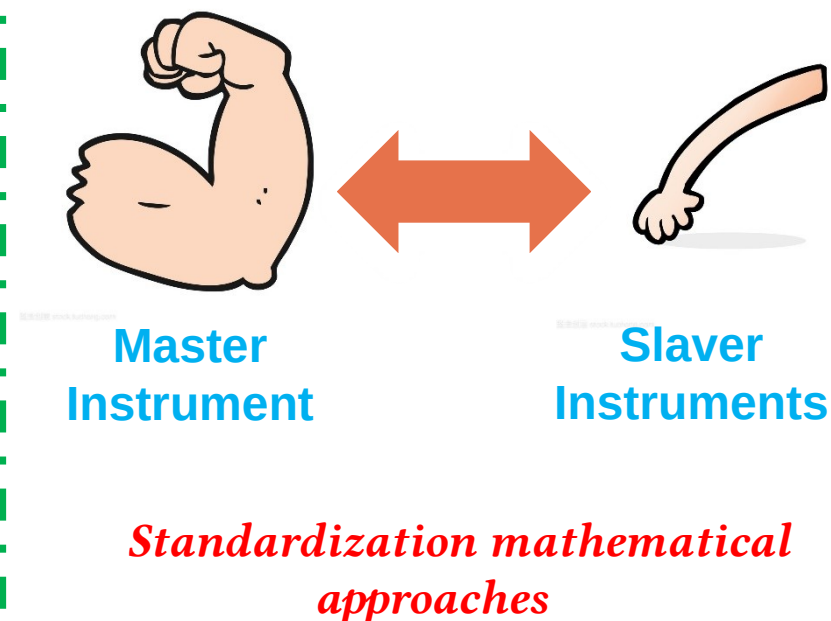
$H_0:$

$$t < t_{\alpha, f} \quad \text{Accept}$$

Improve model stability or Modify the model



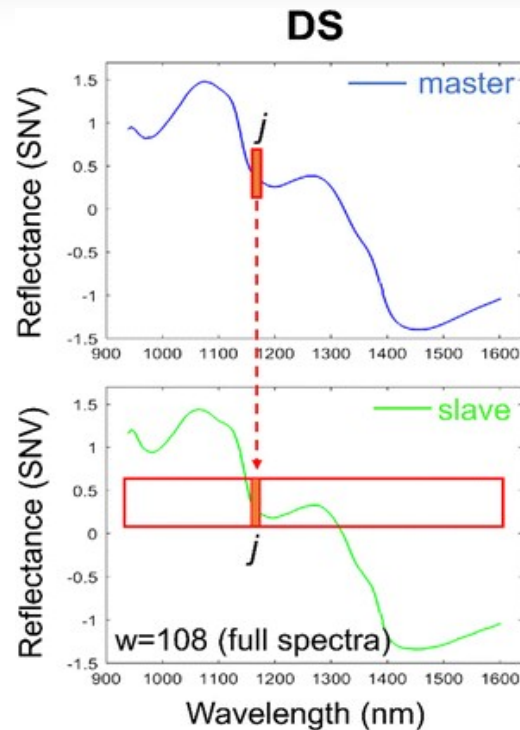
Modify the data



X. G. Shao group. *Progress in Chemistry*, **2017**, 92(8):902-910

- Transfer in spectra ()
- Regression coefficient conversion
- Robust calibration

Commonly used methods for converting spectra among different instruments (or under different conditions) include spectral subtraction correction (SSC), Shenk's algorithm, direct standardization (DS), piecewise direct standardization (PDS), etc. These methods usually require a representative set of standard samples (15–30 samples), which are called calibration transfer methods with standard samples.

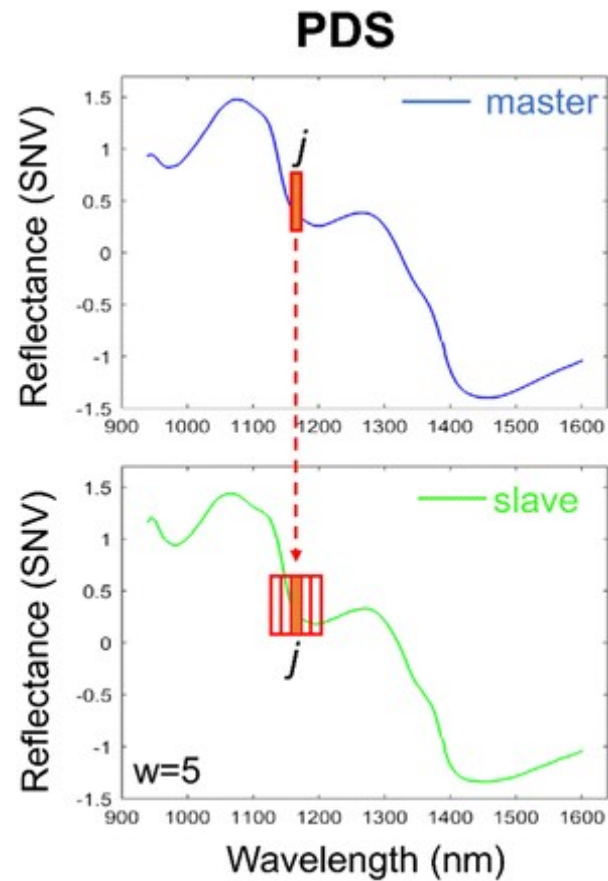


$$X_m = X_s \cdot F \quad \longrightarrow \quad F = X_s^+ X_m$$

- Simple calculation
- Overfitting (sample size $\ll$ Spectral points)
- Moore-Penrose generalized inverse matrix introduces errors (The sample size required is relatively large)

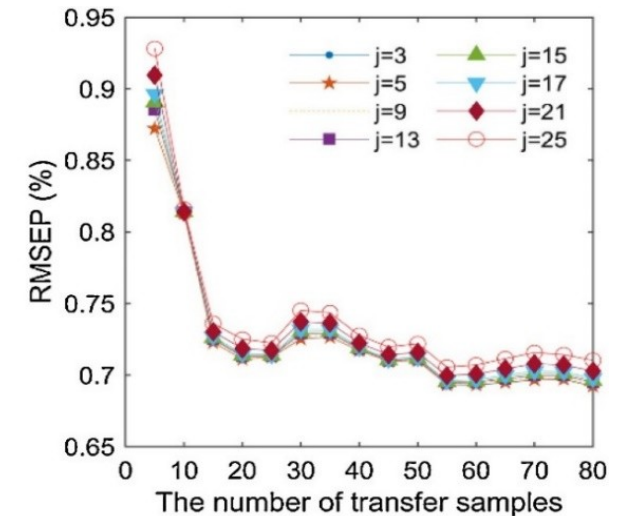
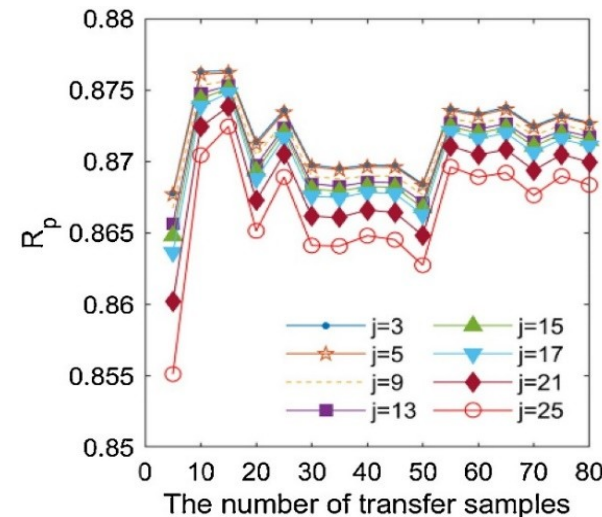
Y. D. Wang et al. *Anal. Chem.* **1991**, 63, 23m 2750-2756
Y. Y. Pu et al. *Food Anal. Methods.* **2018** (11):1021-1033
L. J. Li group. *Postharvest. Biol. Tec.* **2022**, 183, 111720

Piecewise direct standardization PDS

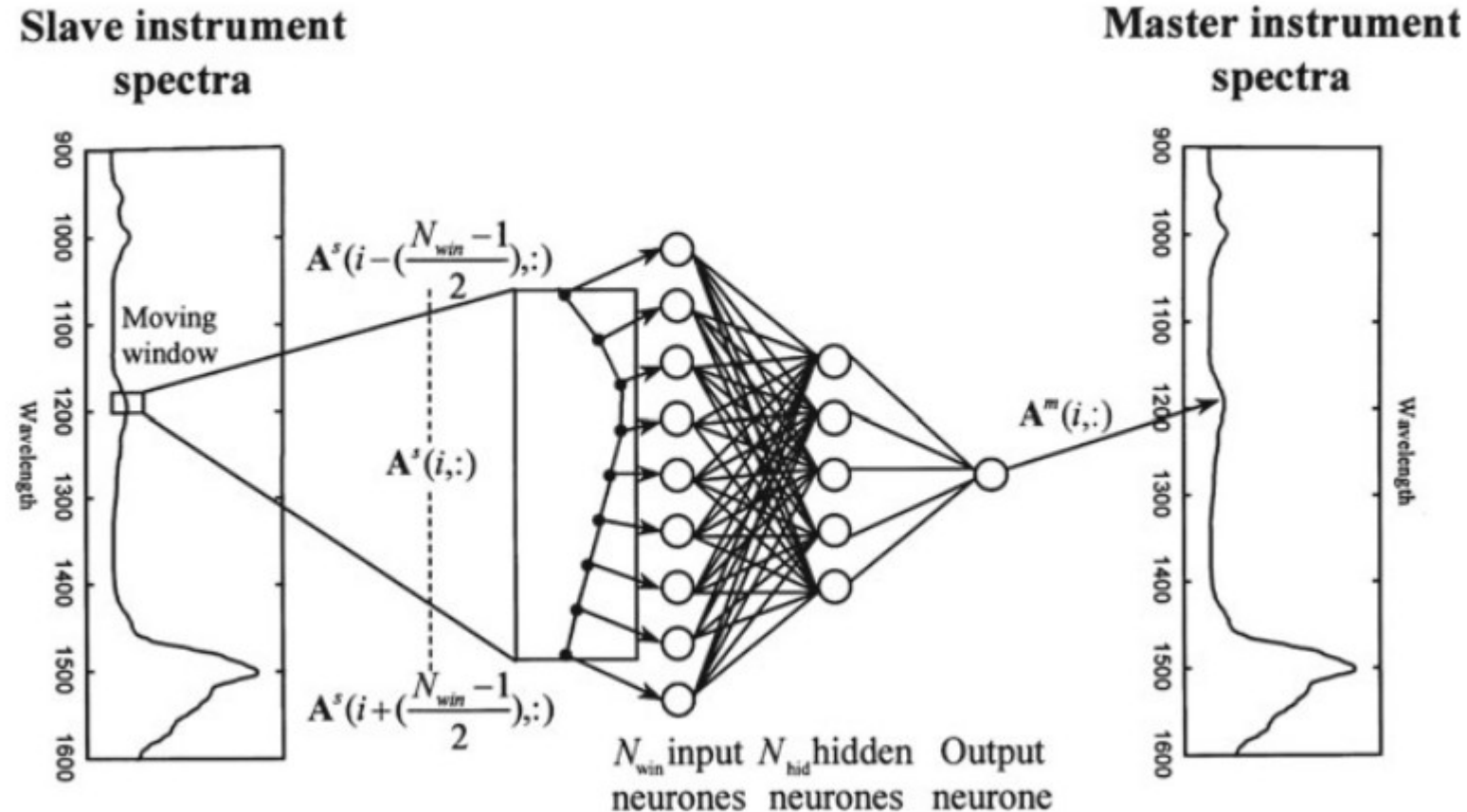


$$S_{m,i} = f_i(S_{S,i-k}, S_{S,i-k+1}, \dots, S_{S,i}, \dots, S_{S,i+w-1}, S_{S,i+w})$$

Principal component regression (PCR)
Partial least squares regression (PLS)

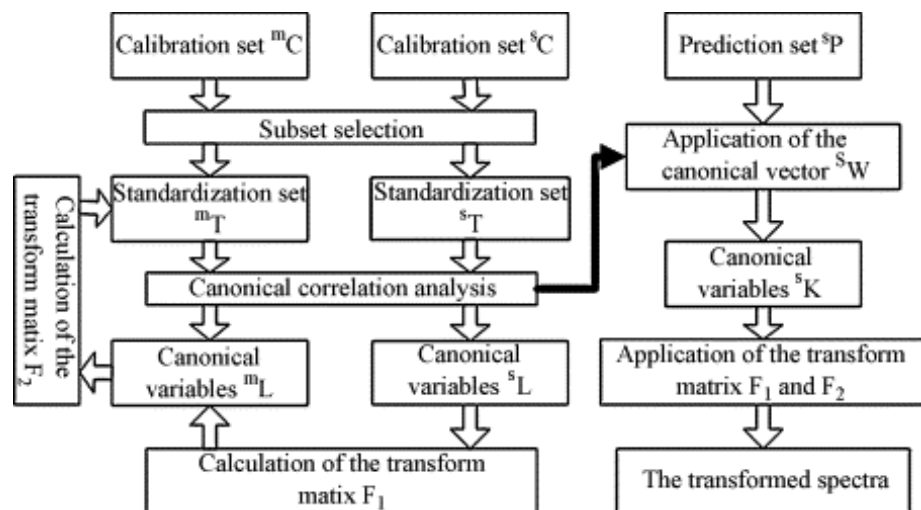


- Sample size decreased;
- Reduce overfitting;
- Window size is difficult to determine



When calculating the transfer matrix F , besides the PLS method, methods such as artificial neural networks can also be used.

Canonical Correlation Analysis (CCA)



- Multiple regression analysis between multivariable groups;
- A certain sample size is required;

The average RMSEP for corn data using different methods (SP1 to MP)							
	RMSEP (PDS)					RMSEP (CCA)	RMSEP (recalibration)
	$w = 1$	$w = 3$	$w = 5$	$w = 7$	$w = 9$		
$N = 5$	0.26	0.22	0.21	0.19	<u>0.19</u>	<u>0.28</u>	0.16
$N = 10$	0.26	0.21	0.21	0.19	0.19	0.24	
$N = 15$	0.26	0.22	0.21	0.19	0.19	0.18	
$N = 20$	0.26	0.22	0.21	0.19	0.19	0.18	
$N = 30$	0.25	0.21	0.21	0.19	<u>0.19</u>	<u>0.15</u>	
Key: w stands for the window size of PDS method and N stands for the number of subset samples.							

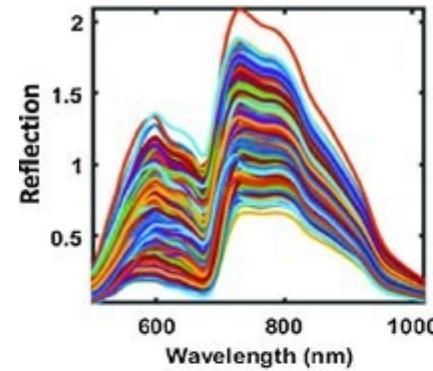
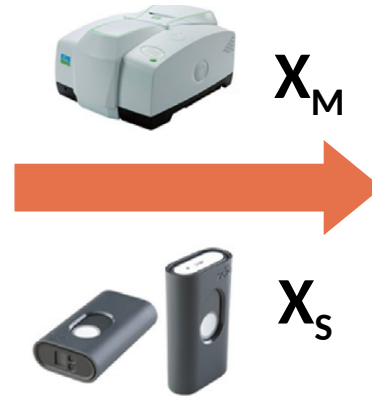
Canonical Correlation Analysis (CCA)

$$\begin{aligned}
 L_m &= X_m W_m & F_1 &= L_s^+ L_m & x_{un}^P &= x_{un} F \\
 L_s &= X_s W_s & F_2 &= L_m^+ X_m & & \\
 & & F &= W_s F_1 F_2 & &
 \end{aligned}$$

Before CCA transfer, Bin et al. used wavelet transform to preprocess the original spectra and performed CCA by wavelet coefficients (WTCCA), which achieved a better transfer effect. Fan et al. calculated the principal components of the master's standard sample spectral matrix X_m by the latent variables of the master PLS model, and then performed spectral transfer (PC-CCA). The validation results were better than the single CCA method.

Similar to CCA, based on spectral regression (SR), Peng et al. decomposed on X_m and X_s , and proposed the spectral regression transfer algorithm. It puts the issue of solving the characteristic function in the regression model, avoids the frequent eigenvalue decomposition process of the dense matrix, and improves the calculation efficiency. Zhang et al. decomposed the spectra based on multi-level simultaneous component analysis (MSCA) and proposed a two-level strategy for spectral transfer algorithm.

Spectral spatial transformation (SST,)

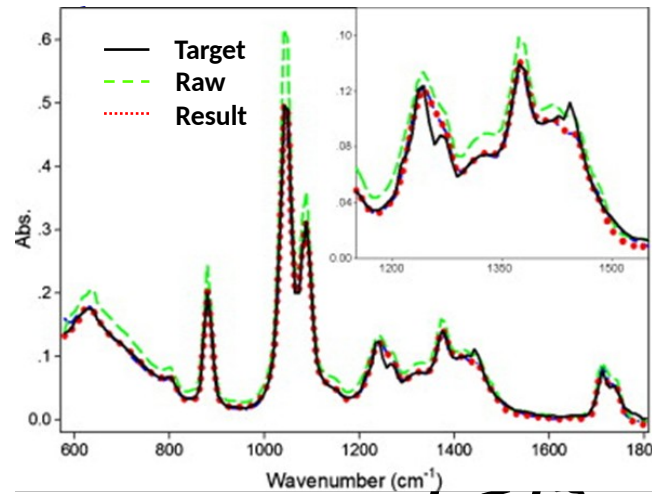


Nutrient Matrix (C)
(Water, vitamins, etc)

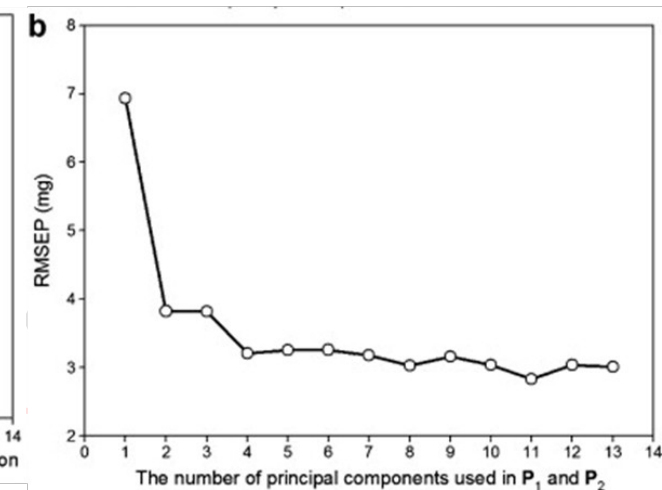
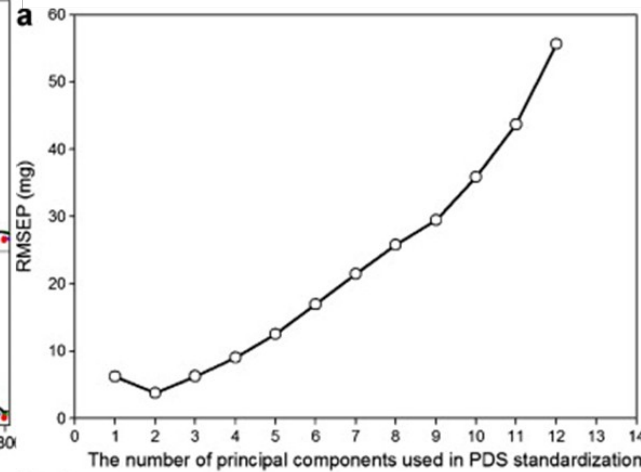
Pure spectrum(P_M, P_S)

Goal: $X_M = X_S F$

Result

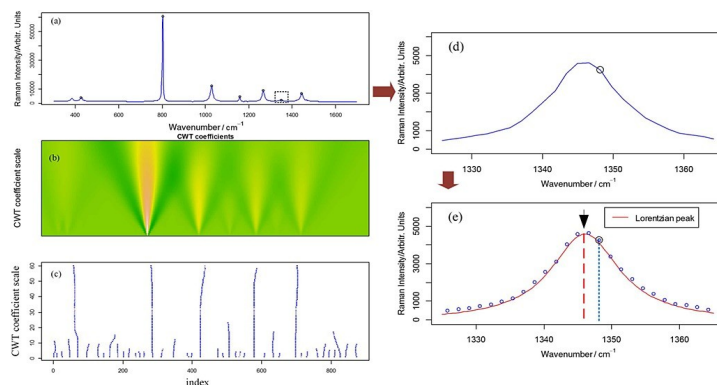


Compared with PDS

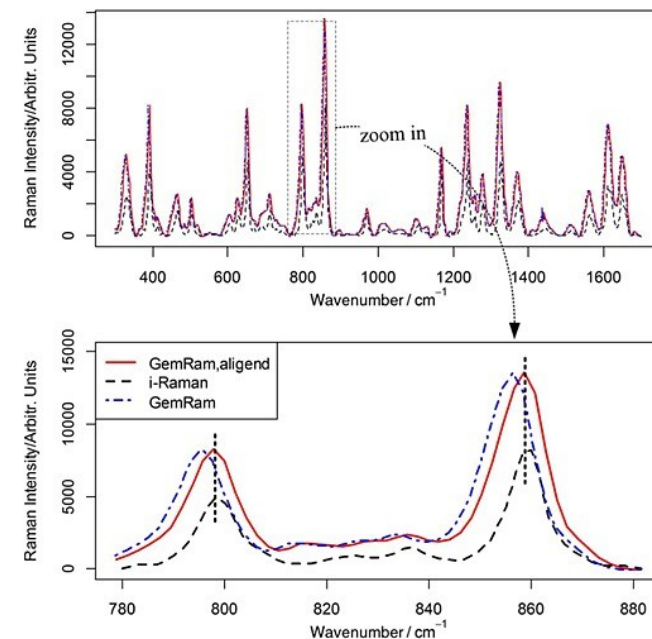
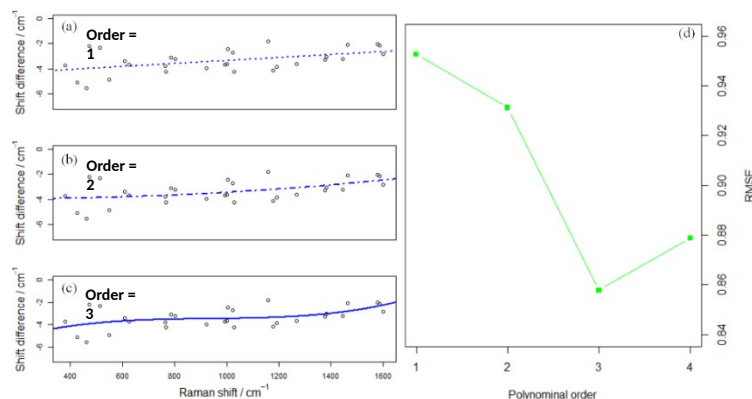


Peak Position Calibration-Raman

Find Peaks



Polyfit



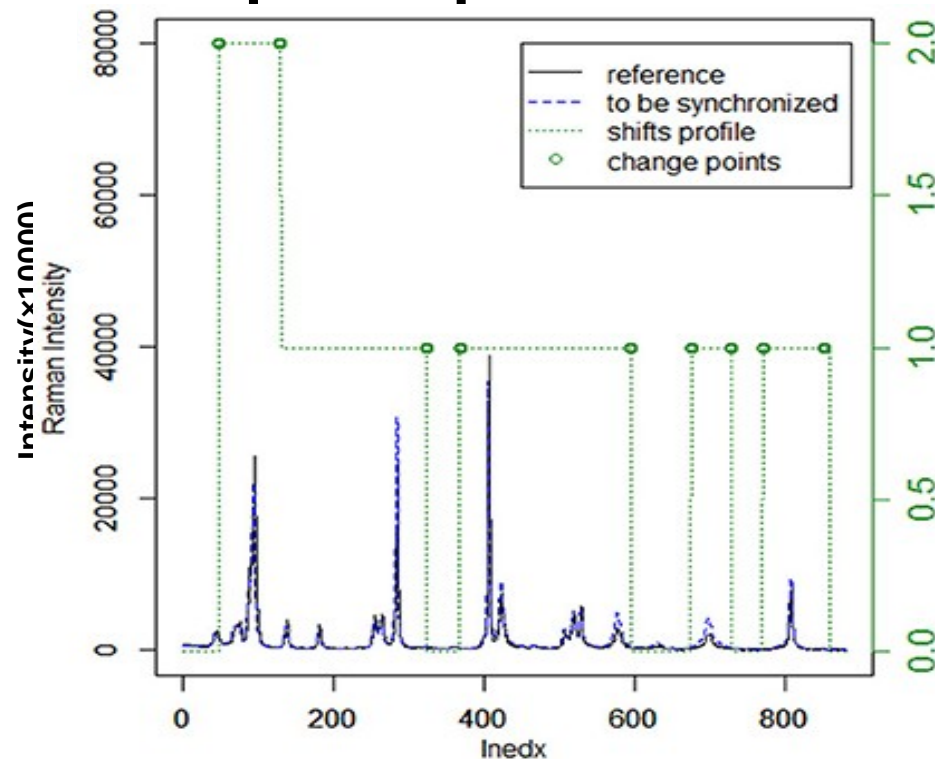
- Move only the peak position
- False positive peak;
- Nonlinear drift

Continuous Wavelet Transform (CWT) + Lorentz fit + Polynomial

Moving Window Fast Fourier Transform (MWFFT) Cross-Correlation

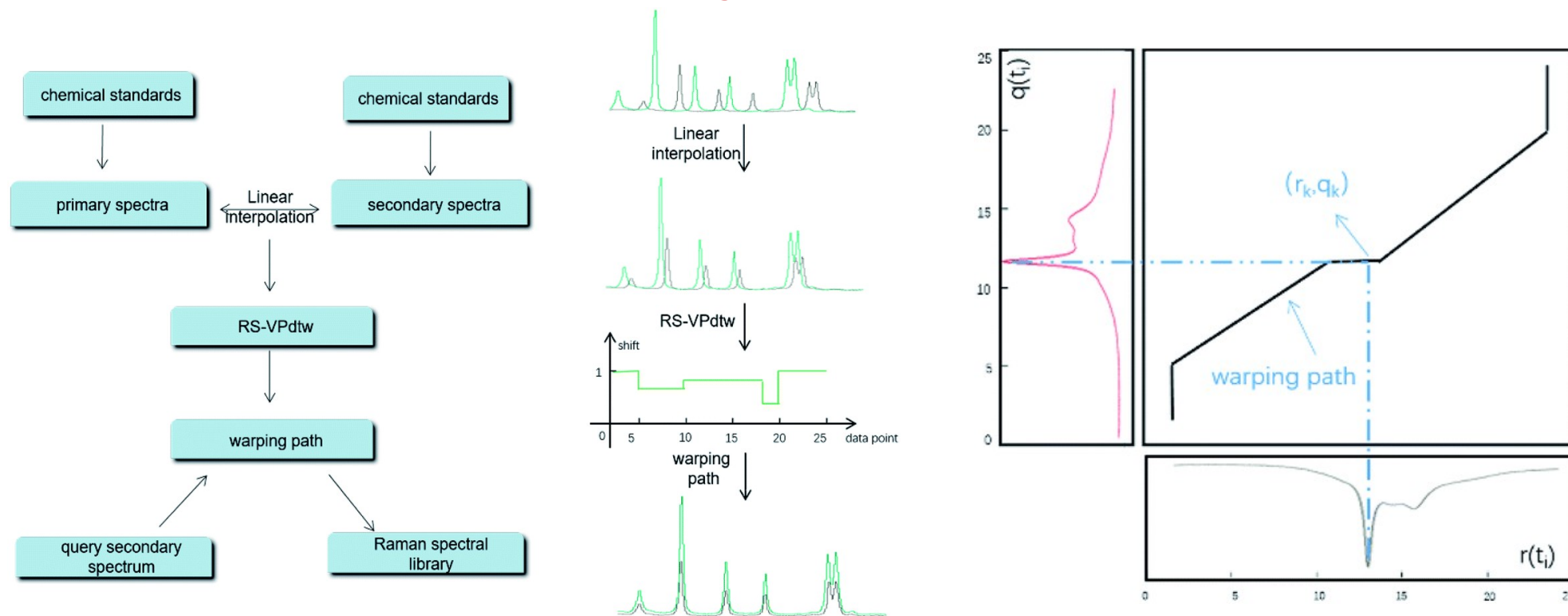
◆ The calibration process

◆ The spectral peak is deformed



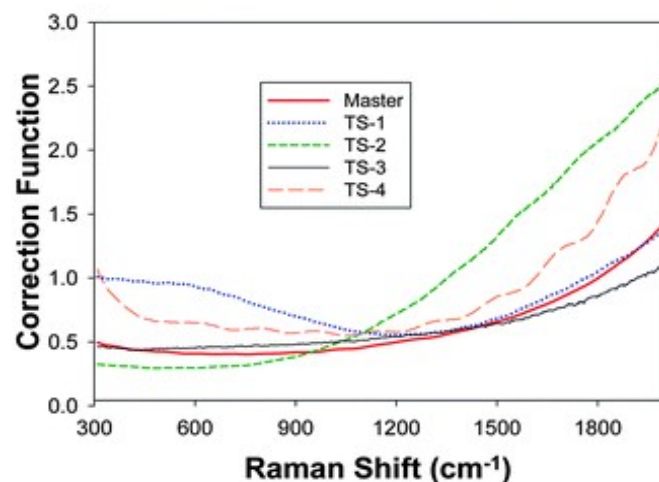
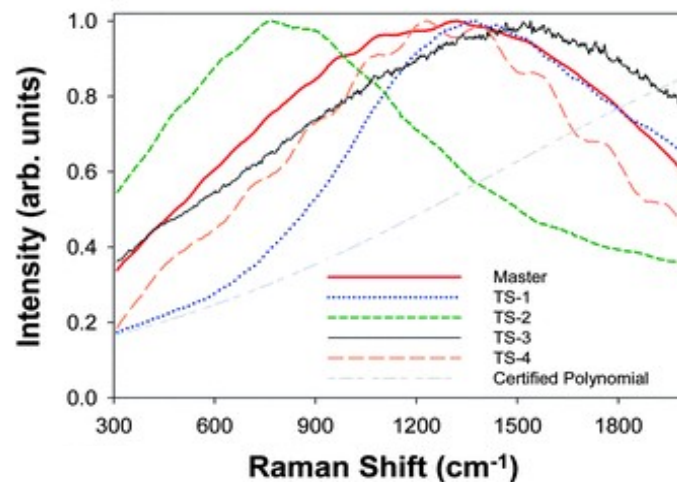
- There is peak deformation;
- The window size will affect the correction result.

Variable Penalty dynamic time warping (RS-VPdtw)



- Only move in one direction;
- The peak deformation is inhibited.

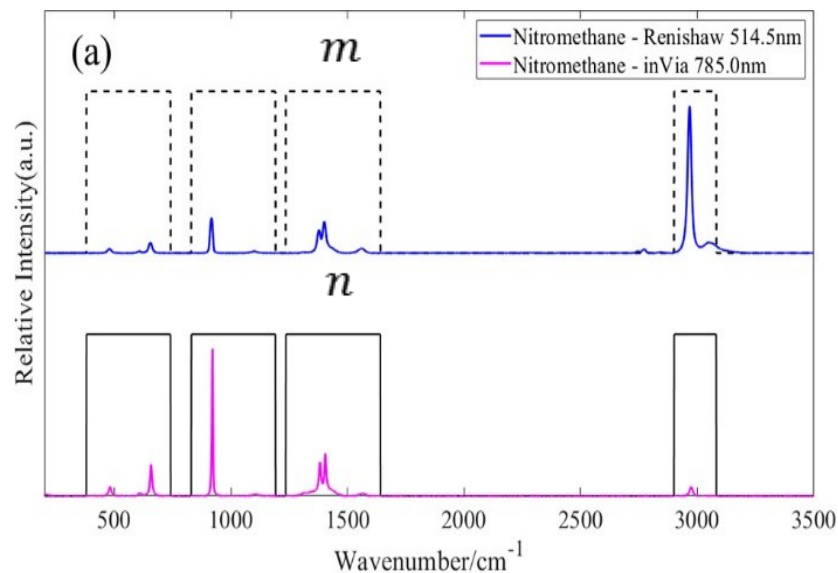
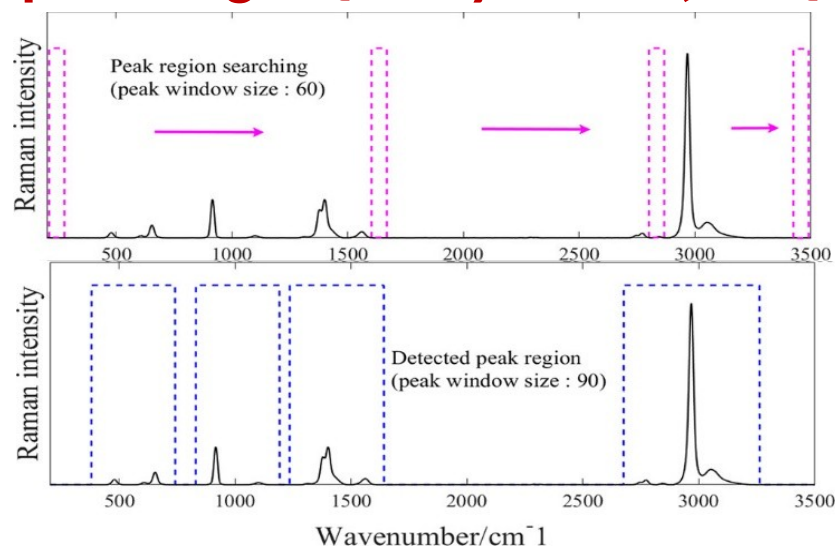
Polynomial



Intensity response standard
standard lamp or standard glass
adjustment instruments are used

- Simple and easy to do
- The response criteria need to be calibrated frequently

Adaptive High Quality Index (AHQI)

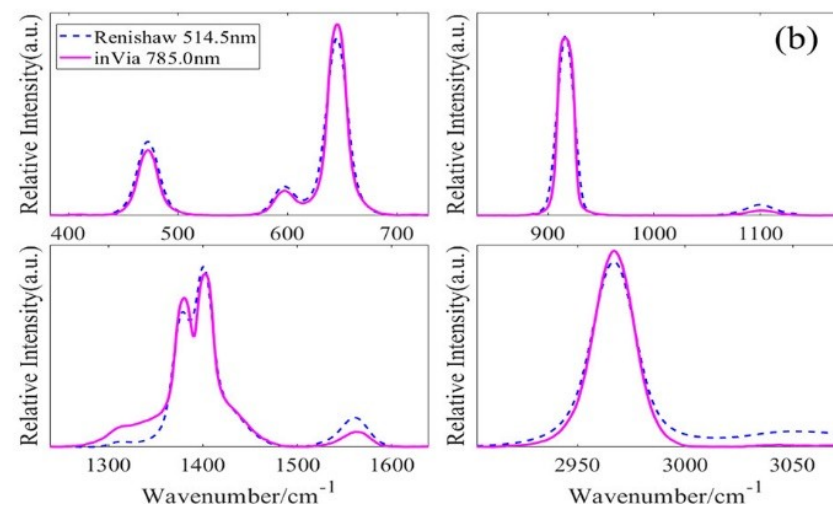


$$AHQI = \frac{S^x + S^y}{2} = \frac{\sum_{i=1}^n w_i^x S_i^x + \sum_{i=1}^m w_i^y S_i^y}{2},$$

$$S^x = \sum_{i=1}^n w_i^x S_i^x; S_i^x = \frac{\|x_i \circ y_i\|}{\|x_i\| \|y_i\|};$$

$$w_i = \frac{1}{m} \text{ 或 } \frac{1}{n};$$

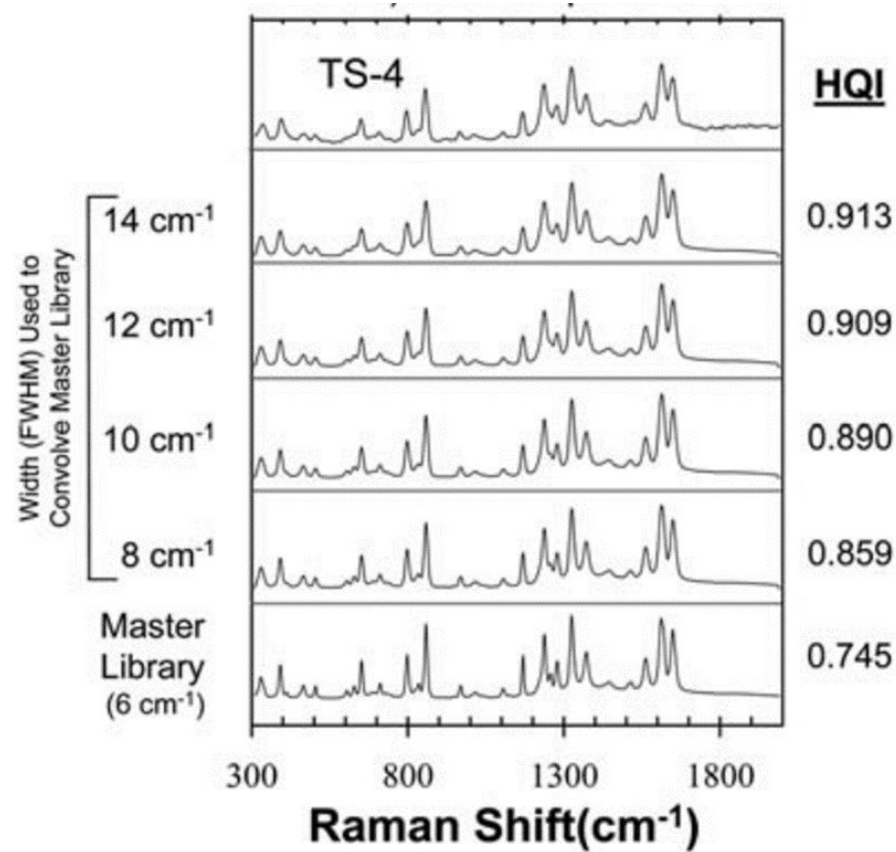
Weights are used for artificial peak adjustment
False relative peak strength variation



Baek S. J. et al. *Anal. Chem.* 2020, 92, 10291-10299

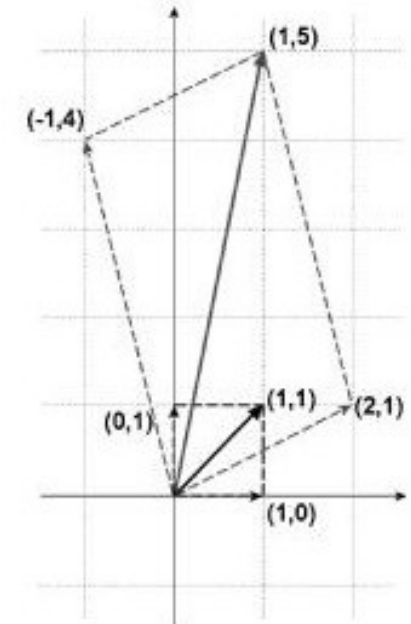
Resolution Calibration

Reduced spectral resolution:
Convolution



Enhanced spectral resolution:
Double Digital Projection Slit (DDPS)

$$\begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



Matrix multiplication is a mapping.

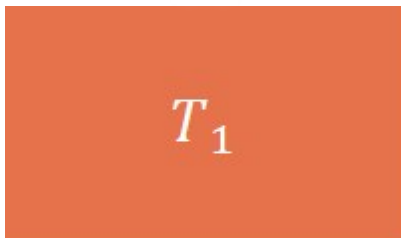
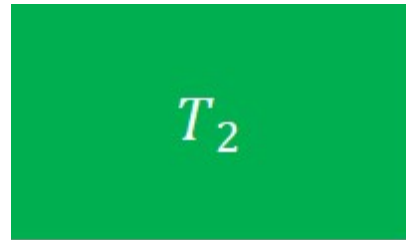
New Algorithms



Resolution Calibration

Double Digital Projection Slit (DDPS)

$$T_2 = H_2 T_0 + n_2$$

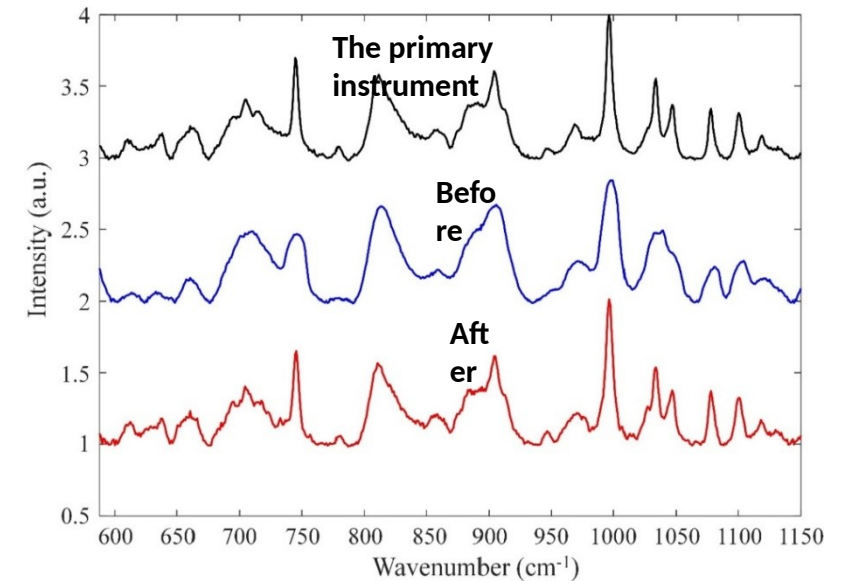


$$T_1 = H_1 T_0 + n_1$$

H – Contains
resolution information
 n – Noise

$$\rightarrow H_1 T_2 = H_1 (H_2 T_0 + n_2)$$

$$\rightarrow H_2 T_1 = H_2 (H_1 T_0 + n_1)$$



- It can improve the spectral resolution
- 150 samples were used as the training set;

Global Calibration, Robust Calibration, and Model Update

Three keywords of global calibration, robust calibration, and model update (or model maintenance) are often referred to confusingly in spectral multivariate calibration analysis, and actually they have a lot in common. The **global model calibration**, also known as the augmented hybrid calibration, hybrid calibration, spiking method in the literature, usually refers to the expansion of the calibration set of the master by adding spectra under different test conditions, spectra measured from different instruments, etc. to build a model so as to achieve model sharing under different instruments, different measurement environments, or different sample types.

Robust calibration often refers to the establishment of a model that is not sensitive to external influence factors through spectral preprocessing algorithms, wavelength variable screening algorithms, etc. Therefore, it can be considered that the establishment of a global calibration is a means to achieve a robust model, and both two methods can also be combined to establish a robust global model. However, the **model updating or calibration maintenance** covers a broader range. In the traditional concept, when encountering samples outside of the model (chemical composition or physical state of the sample changes) or the instrument is aging over time, what we need is model update or model maintenance. In general concept, the process of establishing a global model or a robust calibration actually also belongs to the category of model update or model maintenance. Thus, global calibration, robust calibration, and model updates (or model maintenance) are usually employed to solve calibration transfer problems.

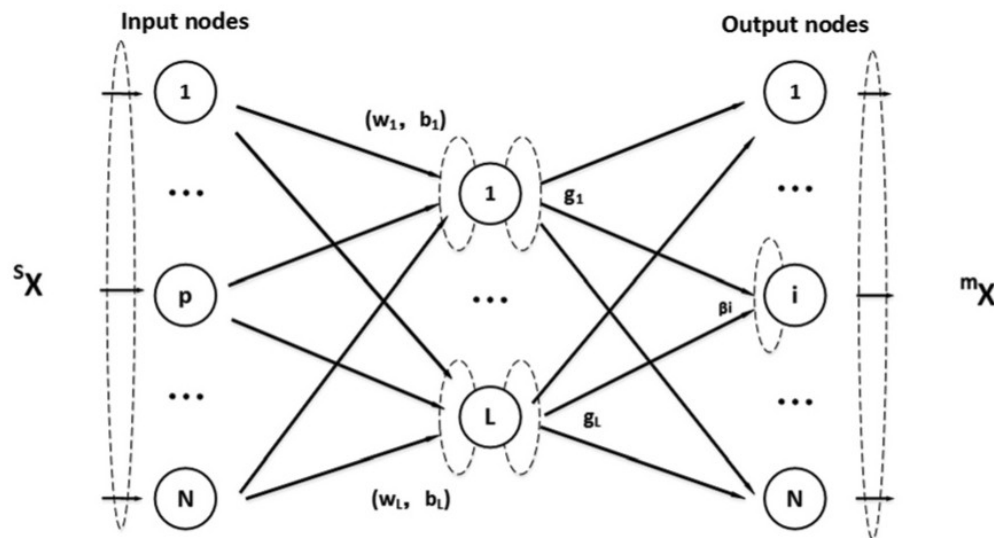


3. AI for data standardization

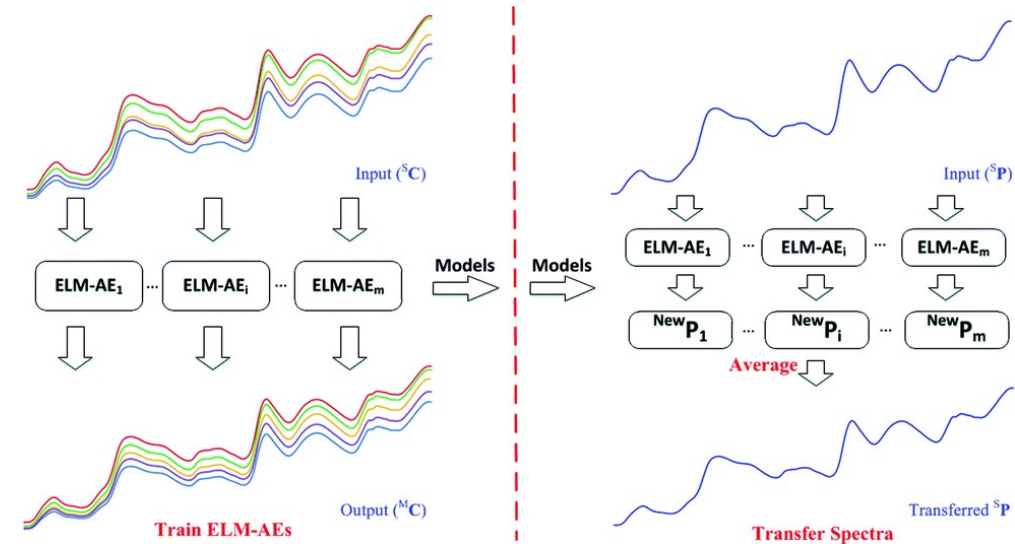
Extreme Learning Machine Auto-Encoder Method (TEAM) - NIR



The network structure of ELM-AE
for calibration transfer

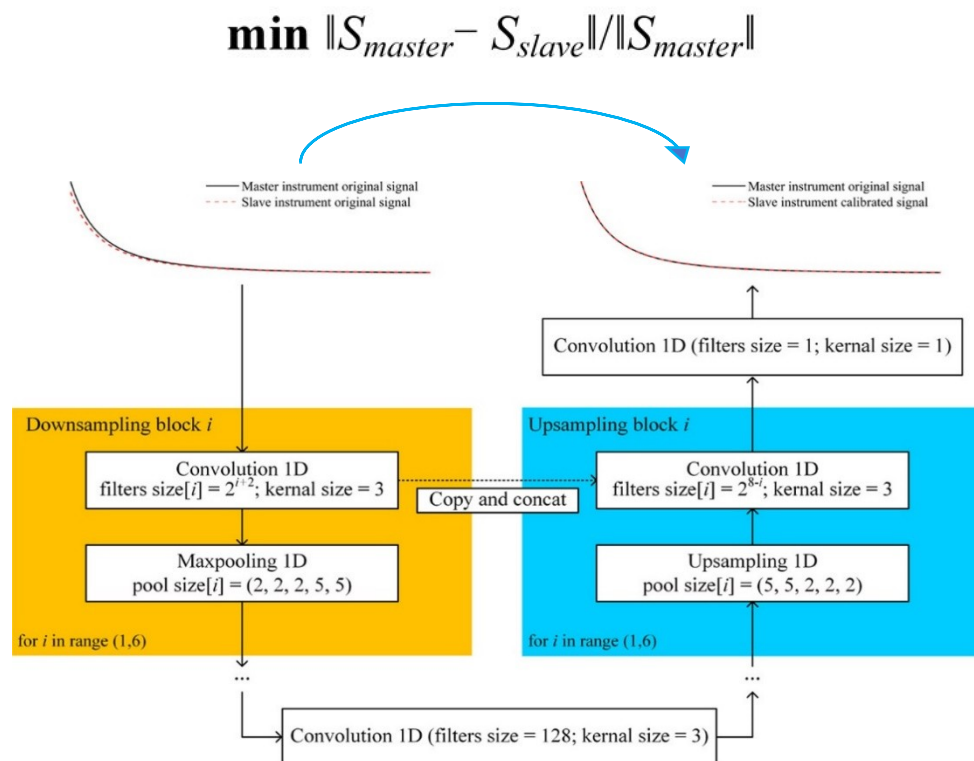


Ensemble Extreme Learning
Machines (EELM)

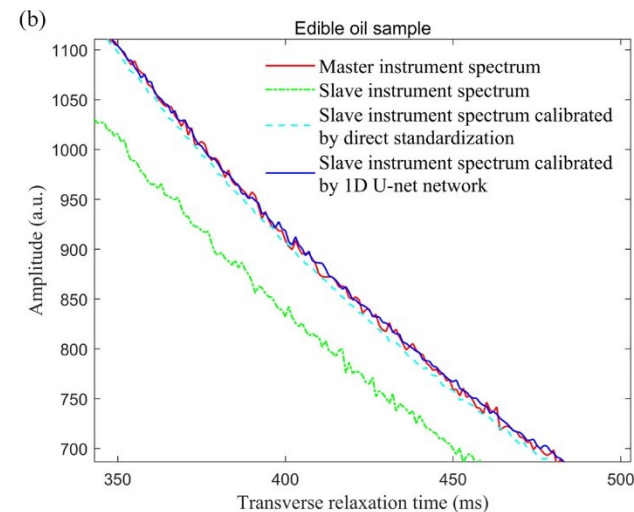


- Only one hyperparameter [number of hidden layer nodes];
- Model training requires relatively little data ~50

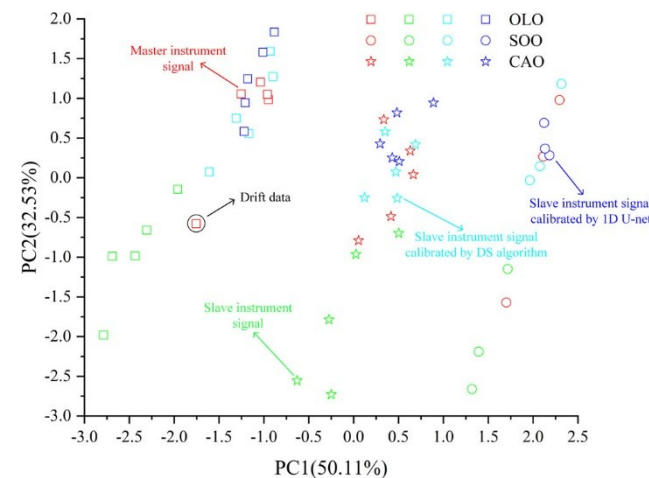
1D U-Net (NMR)



- It's better than direct standardization
- Spectral, Principal component map, Shift
- Number of training sets 91;



Spectral Response



Principal component map

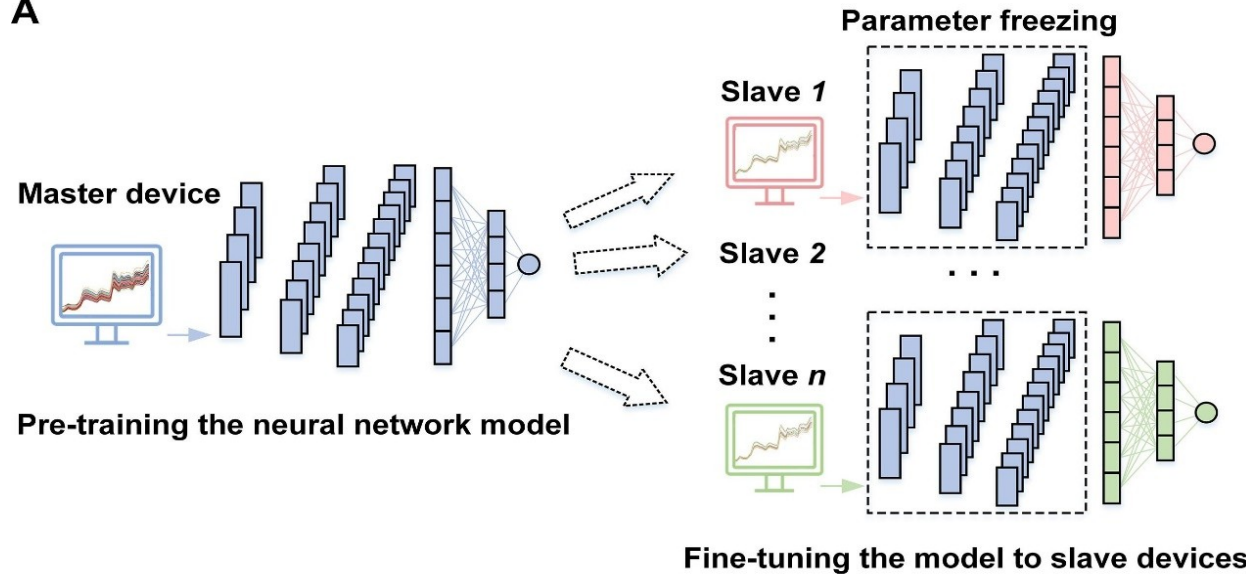
X. W. Hou et al. *Anal. Chem.* **2021**, 93,30, 10469-10476

Transfer Learning



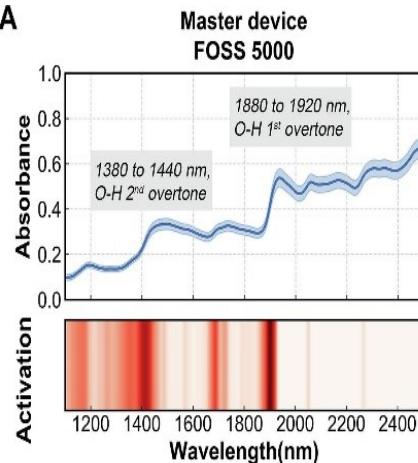
DeepTranSpectra (DTS)

A

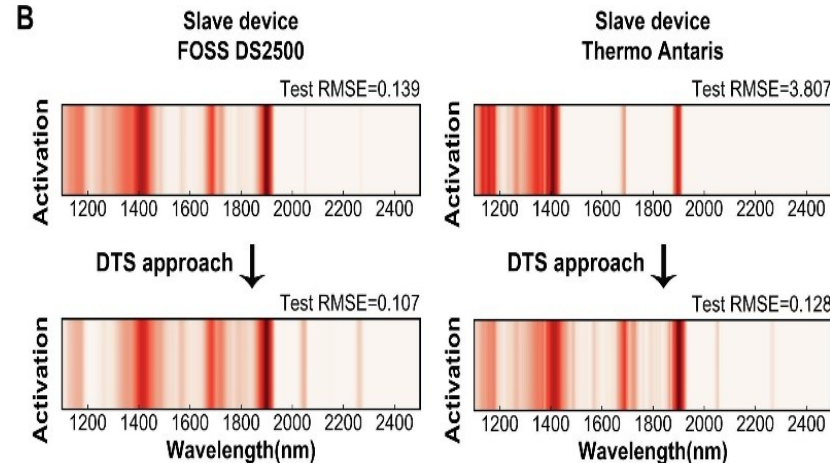


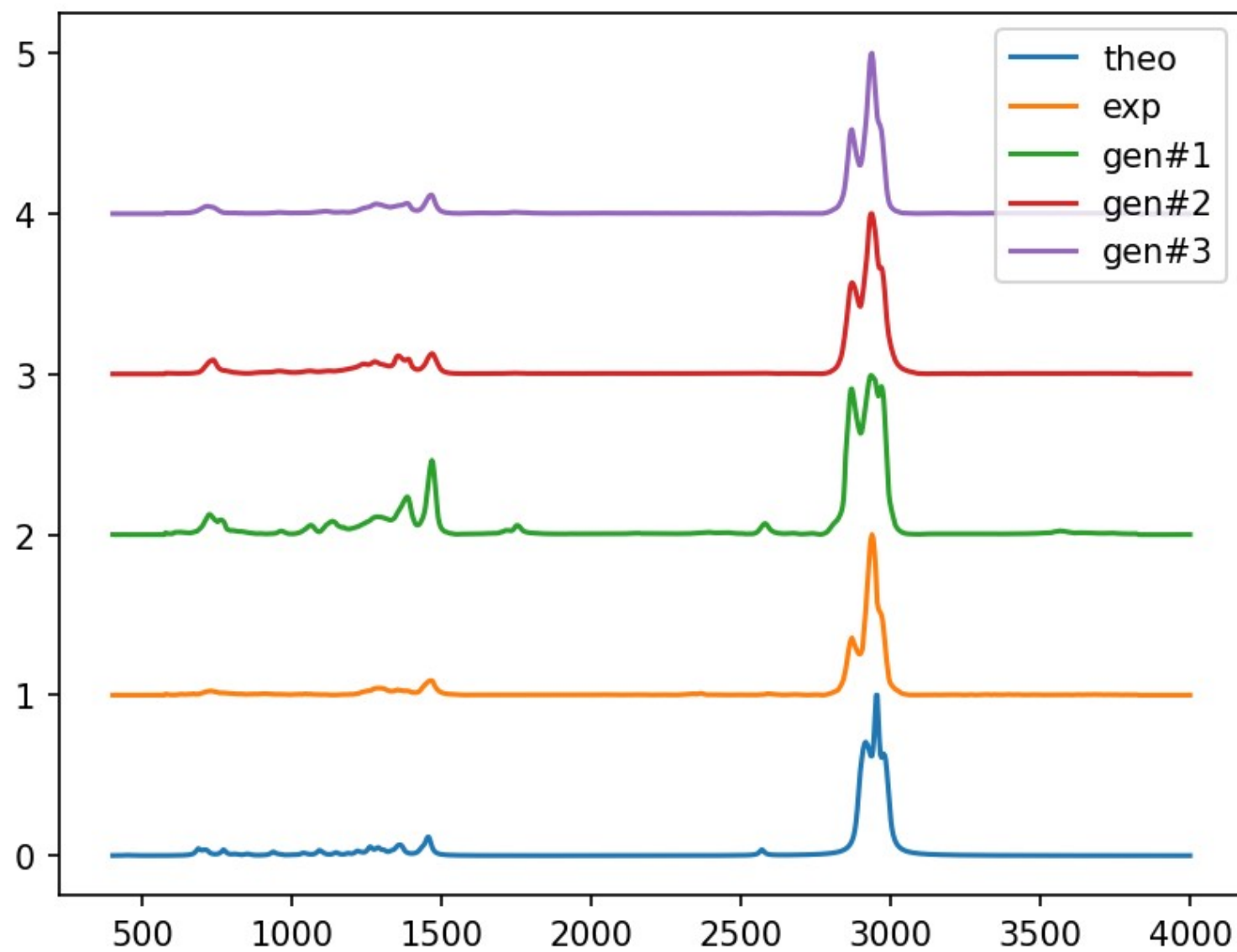
- Modify only the parameters of the full connection layer;
- Transfer learning is used for model transfer;
- The characteristics of activation are similar after migration;

A



B





To be continued.....